Supporting Information: title to be chosen

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Methods

Baselines

> **Plan**
>
> - `randomwalk` and `randomwalknowcast` in full.
> - Key references: (Johnson et al. 2025).

Foundation model setup and fine-tuning

> **Plan**
>
> - How Chronos-2 and LoRA work, in brief, with schematics.
> - Foundation model setup and fine-tuning settings.
> - Key references: (Ansari et al. 2025; Hu et al. 2021).

Synthetic data generation

> **Plan**
>
> - Synthetic data generation in full, with an overview of the epinowcast model used.
> - Key references: (Abbott et al. 2026; Lison et al. 2024; Park et al. 2024; Abbott 2026).

The epinowcast model used to simulate outbreaks

We simulated each synthetic outbreak from the generative model of epinowcast (**placeholder?**). We used `enw_simulate()`, which runs the epinowcast Stan model with fixed parameters and no likelihood. It draws reported counts from the model's generated quantities, so the simulated data come from the same model epinowcast fits. `enw_simulate()` is on the `feature/829-simulate-forecast` branch of epinowcast, at commit `e084b08089796ca7def8dac43fd62d29ae34258b`. The simulation script is `scripts/finetune_simulate.R`, with the grid named r2-cal5, and the settings drawn for each outbreak are in the synthetic scenario grid table. Each outbreak first draws its design, the categorical choices below, and then its continuous settings. Outbreaks are simulated on a daily time step by stratum, with a longest reporting delay of 3 to 12 weeks, 5 to 10 times the delay mean. A parameter we did not set is drawn from its epinowcast prior.

Strata

An outbreak has one stratum, 2 to 26 places, 2 to 8 age groups in one place, or 6 to 16 places by 3 to 7 age groups. Strata share part of their growth through a weighted sum of a shared path and a path of their own, and places by ages also share a path by place. Some places are quiet, seeded with almost no cases. The initial latent cases (`expr_lelatent_int`) are a drawn total split over strata by log-normal weights.

Expectation model

Expected cases come from `enw_expectation()`, driven by a daily growth path given to `enw_simulate()` as `growth_rate`. In the two renewal processes the path is log R_t, and expected latent cases are $\lambda_t = R_t \sum_{s=1}^{45} g_s \lambda_{t-s}$ with a discretised gamma generation time g_s. A log R_t path starts at a drawn value, has a Poisson number of step changes and a daily random walk, and is held between -1.2 and 1.2. One renewal process adds an incubation period, a discretised gamma delay from latent cases to onsets (`latent_reporting_delay`). The other four processes have no generation time, and the path is the daily growth rate of log incidence. It is a weekly random walk, white noise (a random walk on log incidence), one to three step changes, or a smooth sine wave, held between -0.06 and 0.06 a day.

Reference date model

The reporting delay of an onset is set by `enw_reference()`. It is a parametric delay alone, a non-parametric hazard alone, or both. The parametric delay is a lognormal, gamma or exponential, with a drawn mean and coefficient of variation. The non-parametric hazard has random effects by delay, by onset week, or both, on the logit hazard of being reported. With no parametric delay, the hazard's intercept gives a geometric delay with the drawn mean. Strata have random effects on the delay. One outbreak in five has a stable delay. In the others the delay may drift linearly in its location and spread, step up to three times at random weeks, and follow a weekly random walk (`rw(week)`).

Report date model

Effects of the report date are set by `enw_report()`. They may include random effects by day of the week of report, a `holiday` covariate that lowers reporting on a few dates, and report week effects, `(1 | week)` or `rw(week)`. In half the outbreaks reports are made every day. In the rest, a structural table from `enw_structural_reporting_metadata()`, given to `enw_report(structural)`, limits reports to one to three weekdays, drawn from Monday to Friday. The report days may change once, at a drawn date.

Observation model

The count of cases with onset on day t reported with delay d is negative binomial (NB2 or NB1) or Poisson, set by `enw_obs(family)`. Its mean is the expected onsets on day t times the probability of a delay of d given the reference and report date models. The overdispersion `sqrt_phi` is drawn on the NB2 scale, and the NB1 dispersion is scaled to match it (epinowcast issue 937). A draw that fails or gives an outbreak outside its size range is drawn again with the same design, up to five times.

Table 1: The simulation grid (r2-cal5), with the reason for each range. Ranges are chosen to be plausible across infectious diseases rather than for one disease.

Setting	Range or mix	Reason
Strata	one 30%, 2-26 places 40%, 2-8 age groups 20%, 6-16 places by 3-7 ages 10%	Single series, regional, age-structured and nested surveillance all occur.
Quiet strata	up to 90% of places seeded with almost no cases	Many places report no cases in a local outbreak.
Generative process	renewal, renewal with incubation, growth walk, log-incidence walk, growth steps, smooth growth; a sixth each	Different shapes of transmission and reporting of infection.
Generation time	gamma, mean 2-20 days, CV 0.3-0.7	Spans fast respiratory to slow haemorrhagic-fever transmission.
Starting log Rt	normal, mean 0.1, sd 0.2	Reproduction numbers mostly 0.7 to 1.8 at the start.
Starting daily growth (no generation time)	normal, mean 0.01, sd 0.02; limits plus or minus 0.1	Doubling or halving times of a week or longer.
Trajectory	turning over, flat, rising then falling, second wave; a quarter each	Forecast origins fall in rising, peak, falling and resurging phases.
Seed	0.2-200 a day, log-uniform	From single introductions to established spread.
Size	10 to 300,000 cases times strata/26; draws expected over 1.5 times the cap rejected before simulating	From small outbreaks to national-scale waves.
Duration	16-40 weeks (16-24 for places by ages)	One season or wave.
Delay distribution	lognormal, gamma (CV 0.7 or more), exponential, or a non-parametric hazard alone	Common parametric and free forms.
Delay mean	2-21 days, log-uniform	From near-real-time reporting to slow confirmation.
Longest delay	3-12 weeks, 5-10 times the delay mean	Realistic tails without the numerical failure of very long tails (epinowcast #939).

Setting	Range or mix	Reason
Delay changes	stable 20%; otherwise drift, 0-3 step changes, spread changes, weekly walk, week hazard effects	Reporting systems change.
Report-date effects	day of week 70%, holidays 30%, report-week effects 40%, weekday-only reporting 50%, weekday switch 30%	Common reporting patterns.
Observation model	negative binomial NB2 40%, NB1 40% (dispersion matched to NB2, epinowcast #937), Poisson 20%	From Poisson to heavily overdispersed counts.
Overdispersion	sqrt(phi) 0.05-1.5	From near Poisson to very noisy counts.

The grid has two limitations. Simulated strata are less sparse than BDBV's provinces: the share of stratum-weeks with no cases reaches 0.75, against 0.89 for BDBV. Delays stop at 12 weeks, while the German data are capped at 80 days by their reporting, so their 99% delay point (81 days) is beyond the grid.

The held-out synthetic set

The held-out set is 100 outbreaks simulated with `scripts/finetune_simulate.R` on the r2-cal5 grid, 25 for each of the seeds 9001, 9002, 9003 and 9004. The fine-tune never used these seeds, so the held-out outbreaks are new draws from the grid it was trained on. If an outbreak's design failed five times and was skipped, a further seed tops the set up to 100. They are written to `output/finetune/heldout`, as `counts-<seed>.csv.gz` and `settings-<seed>.csv`. The evaluation workflow (`paper/workflow/Snakefile`) runs the models on the set in its `paper` target only.

The held-out set was reserved for the final evaluation. No model was scored on it during development, and no forecast or score from it was looked at. One early run of the pipeline forecast three held-out outbreaks before this rule was set, and its outputs were deleted unread. We checked the pipeline end to end on a separate validation set of five outbreaks with seed 7001, in `output/finetune/validation`.

The German benchmark is split the same way. Its nowcast dates from 2022-02-10 are held out, and development runs read no forecast from them. Development runs read the German data only up to the release D = 40 days after the last development date, 2022-03-21, which the truth of that date needs. The evaluation workflow checks both, and fails a development run that reads later data.

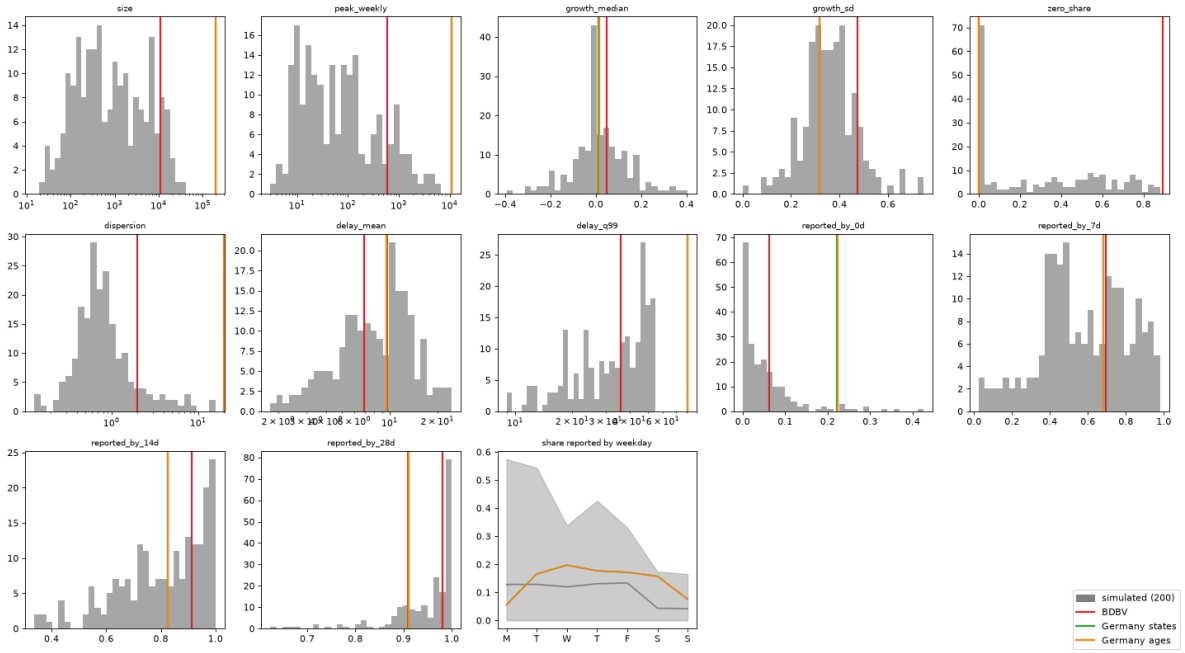


Figure 1: How 200 simulated outbreaks (grey histograms) compare with the BDBV data (red) and the German data by state (green) and by age (orange): outbreak size, weekly peak, growth, the share of stratum-weeks with no cases, dispersion, the mean and 99% point of the reporting delay, the share reported within 0, 7, 14 and 28 days, and the share reported on each weekday.

Protocol

Each outbreak is released weekly, on one weekday from Wednesday to Saturday drawn from its id, as for the fine-tune's training blocks. A release's cut-off is its origin date. Each outbreak has four origins, spread evenly from its sixth release to the last release at least its longest delay (3 to 12 weeks) plus 5 weeks before its last report. Each forecast covers eight weeks of onsets, from four weeks before the week of the cut-off to three weeks after it. Every model forecasts weekly totals by onset week, for the country and each place, with 200 samples (`n_samples` in the workflow config). The Chronos-2 models keep their own release lag of 4 weeks (weekly) or 28 days (daily). The scores use rolling truth, the count reported D days after the end of a target week, with D the outbreak's own 98% reporting delay in whole days. The final simulated count is kept as well. Scores are relative to `randomwalknowcast`.

Correlated trajectories

Plan

- Correlated trajectories: the copula, its rolling-window estimator and the window length.
- Joint block or level by level for the foundation models (#222).

Results to come

Slots for the copula's window length (#221) and for the joint block against level by level (#222).

Scoring

Plan

- Scoring, and its check against scoringutils.
- Key references: (Bosse et al. 2022; Jordan, Krüger, and Lerch 2019; Zanetta and Allen 2024).

Implementation

> **Plan**
>
> - The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
> - The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

> **Plan**
>
> - Score breakdowns by location and horizon for each case study.
> - WIS over time for each case study.
> - Held-out synthetic scores by number of fine-tuning outbreaks.

Held-out synthetic data

Table 2: Scenario settings realised over the evaluated outbreaks: the smallest, median and largest value of each. Illustrative, dummy data

factor	min	median	max	n_locations
delay_mean	2.17	4.31	7.71	8
delay_q98	11	19.5	62	8
gt_mean	6.2	15.1	19.1	8
log_r0	-0.49	0.215	0.89	8
n_places	1	2.5	12	8
report_days	5	5	7	8

Table 3: Held-out synthetic data scores by horizon. Illustrative, dummy data

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	epinowcast	crps	-28	5.4	0.0531	0.216	0.504	0.876	129	0.689	0.748

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	epinowcast	crps	-21	5.17	0.0502	0.192	0.563	0.859	135	0.781	0.71
synthetic	epinowcast	crps	-14	5.52	0.0598	0.195	0.518	0.885	139	0.711	0.802
synthetic	epinowcast	crps	-7	6.55	0.0682	0.18	0.493	0.9	140	0.737	0.774
synthetic	epinowcast	crps	0	12.6	0.121	0.0108	0.489	0.827	139	0.67	0.68
synthetic	epinowcast	crps	7	13.8	0.149	0.125	0.542	0.947	131	0.53	0.624
synthetic	epinowcast	crps	14	24.5	0.202	0.122	0.561	0.902	123	0.566	0.612
synthetic	epinowcast	crps	21	31.1	0.287	0.101	0.513	0.809	115	0.595	0.709
synthetic	fm_complete	crps	-28	10.1	0.0856	-0.381	0.395	0.775	129	1.29	1.21
synthetic	fm_complete	crps	-21	9.1	0.0953	-0.482	0.385	0.77	135	1.37	1.35
synthetic	fm_complete	crps	-14	9.72	0.0964	-0.342	0.381	0.827	139	1.25	1.29
synthetic	fm_complete	crps	-7	11.5	0.112	-0.273	0.414	0.829	140	1.29	1.27
synthetic	fm_complete	crps	0	22.4	0.199	-0.0993	0.446	0.813	139	1.2	1.12
synthetic	fm_complete	crps	7	30.2	0.308	-0.154	0.427	0.847	131	1.16	1.29
synthetic	fm_complete	crps	14	48.3	0.406	0.0666	0.447	0.829	123	1.11	1.23
synthetic	fm_complete	crps	21	42.8	0.444	-0.11	0.53	0.887	115	0.819	1.09
synthetic	fm_ragged	crps	-28	11.6	0.0831	0.556	0.302	0.674	129	1.48	1.17
synthetic	fm_ragged	crps	-21	13.8	0.109	0.63	0.252	0.533	135	2.09	1.55
synthetic	fm_ragged	crps	-14	13.2	0.115	0.584	0.259	0.547	139	1.7	1.54
synthetic	fm_ragged	crps	-7	12.9	0.114	0.383	0.35	0.657	140	1.46	1.29
synthetic	fm_ragged	crps	0	17.7	0.172	0.282	0.396	0.748	139	0.941	0.969

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	fm_chronos	daily_ragged	7	28.6	0.232	0.191	0.389	0.786	131	1.1	0.974
synthetic	fm_chronos	daily_ragged	14	39.8	0.321	0.217	0.39	0.724	123	0.919	0.973
synthetic	fm_chronos	daily_ragged	21	56.7	0.395	0.0441	0.348	0.765	115	1.08	0.974
synthetic	fm_chronos	daily_ragged_ft	28	53	0.0519	0.166	0.535	0.884	129	0.676	0.732
synthetic	fm_chronos	daily_ragged_ft	21	53.8	0.0569	0.152	0.541	0.874	135	0.813	0.805
synthetic	fm_chronos	daily_ragged_ft	14	50.9	0.0581	0.107	0.525	0.885	139	0.772	0.778
synthetic	fm_chronos	daily_ragged_ft	7	47.4	0.0798	0.134	0.421	0.879	140	0.792	0.905
synthetic	fm_chronos	daily_ragged_ft	0	10.6	0.115	0.0932	0.518	0.906	139	0.562	0.645
synthetic	fm_chronos	daily_ragged_ft	7	16.5	0.174	0.0362	0.481	0.863	131	0.634	0.729
synthetic	fm_chronos	daily_ragged_ft	14	26.6	0.228	0.0798	0.496	0.902	123	0.613	0.693
synthetic	fm_chronos	daily_ragged_ft	21	33.7	0.305	0.0681	0.435	0.861	115	0.645	0.752
synthetic	fm_chronos	weekly_complete	28	10.8	0.102	-0.529	0.302	0.713	129	1.38	1.44
synthetic	fm_chronos	weekly_complete	21	10.6	0.102	-0.483	0.385	0.719	135	1.6	1.45
synthetic	fm_chronos	weekly_complete	14	10.8	0.0993	-0.42	0.432	0.791	139	1.39	1.33
synthetic	fm_chronos	weekly_complete	7	13.7	0.131	-0.268	0.436	0.814	140	1.54	1.48
synthetic	fm_chronos	weekly_complete	0	17.7	0.183	-0.189	0.54	0.878	139	0.941	1.03
synthetic	fm_chronos	weekly_complete	7	30.5	0.285	-0.0632	0.45	0.893	131	1.17	1.19
synthetic	fm_chronos	weekly_complete	14	53.3	0.416	-0.106	0.431	0.846	123	1.23	1.26
synthetic	fm_chronos	weekly_complete	21	51.5	0.47	-0.0654	0.426	0.896	115	0.985	1.16
synthetic	fm_chronos	weekly_ragged	28	47.8	0.0696	-0.215	0.442	0.783	129	0.916	0.98

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	fm_chronos_weekly_ragged	crps	21	6.91	0.0701	-0.0697	0.489	0.8	135	1.05	0.992
synthetic	fm_chronos_weekly_ragged	crps	14	8.15	0.0847	-0.0581	0.446	0.755	139	1.05	1.14
synthetic	fm_chronos_weekly_ragged	crps	7	11.3	0.111	-0.022	0.379	0.764	140	1.27	1.26
synthetic	fm_chronos_weekly_ragged	crps	0	16.3	0.161	0.00683	0.482	0.835	139	0.867	0.907
synthetic	fm_chronos_weekly_ragged	crps	-7	28.6	0.267	0.00565	0.382	0.763	131	1.09	1.12
synthetic	fm_chronos_weekly_ragged	crps	-14	30.8	0.309	0.0122	0.463	0.846	123	0.711	0.936
synthetic	fm_chronos_weekly_ragged	crps	-21	57.2	0.427	0.0567	0.391	0.8	115	1.09	1.05
synthetic	fm_chronos_weekly_ragged_ft	crps	-28	7.14	0.0608	0.315	0.543	0.845	129	0.912	0.857
synthetic	fm_chronos_weekly_ragged_ft	crps	-21	7.51	0.0628	0.342	0.511	0.837	135	1.14	0.888
synthetic	fm_chronos_weekly_ragged_ft	crps	-14	7.79	0.0627	0.254	0.547	0.899	139	1	0.84
synthetic	fm_chronos_weekly_ragged_ft	crps	-7	9.07	0.0775	0.216	0.529	0.921	140	0.897	0.878
synthetic	fm_chronos_weekly_ragged_ft	crps	0	13.5	0.132	0.14	0.576	0.892	139	0.718	0.744
synthetic	fm_chronos_weekly_ragged_ft	crps	-7	23.1	0.176	0.11	0.588	0.947	131	0.884	0.737
synthetic	fm_chronos_weekly_ragged_ft	crps	-14	31.6	0.255	0.0405	0.472	0.886	123	0.73	0.774
synthetic	fm_chronos_weekly_ragged_ft	crps	-21	46.8	0.333	0.14	0.53	0.817	115	0.895	0.82
synthetic	randomwalk	crps	-28	8.52	0.0734	-0.325	0.535	0.922	129	1.09	1.03
synthetic	randomwalk	crps	-21	8.93	0.0787	-0.296	0.504	0.911	135	1.35	1.11
synthetic	randomwalk	crps	-14	9.5	0.088	-0.248	0.518	0.95	139	1.22	1.18
synthetic	randomwalk	crps	-7	9.76	0.102	-0.147	0.55	0.979	140	1.1	1.15
synthetic	randomwalk	crps	0	21.1	0.185	-0.0741	0.576	0.957	139	1.13	1.04

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	randomwalk	crps	7	27.4	0.228	-0.0545	0.71	0.962	131	1.05	0.955
synthetic	randomwalk	crps	14	44.5	0.344	-0.0631	0.593	0.959	123	1.03	1.04
synthetic	randomwalk	crps	21	56.3	0.441	-0.0797	0.617	0.957	115	1.08	1.09
synthetic	randomwalknowcast	crps	-28	7.83	0.071	0.0999	0.612	0.953	129	1	1
synthetic	randomwalknowcast	crps	-21	6.62	0.0707	0.0916	0.637	0.97	135	1	1
synthetic	randomwalknowcast	crps	-14	7.76	0.0746	0.159	0.647	0.971	139	1	1
synthetic	randomwalknowcast	crps	-7	8.88	0.0882	0.0927	0.707	0.971	140	1	1
synthetic	randomwalknowcast	crps	0	18.8	0.178	0.033	0.597	0.957	139	1	1
synthetic	randomwalknowcast	crps	7	26.1	0.239	0.135	0.725	0.969	131	1	1
synthetic	randomwalknowcast	crps	14	43.4	0.33	0.0618	0.675	0.967	123	1	1
synthetic	randomwalknowcast	crps	21	52.3	0.406	-0.0158	0.643	0.974	115	1	1

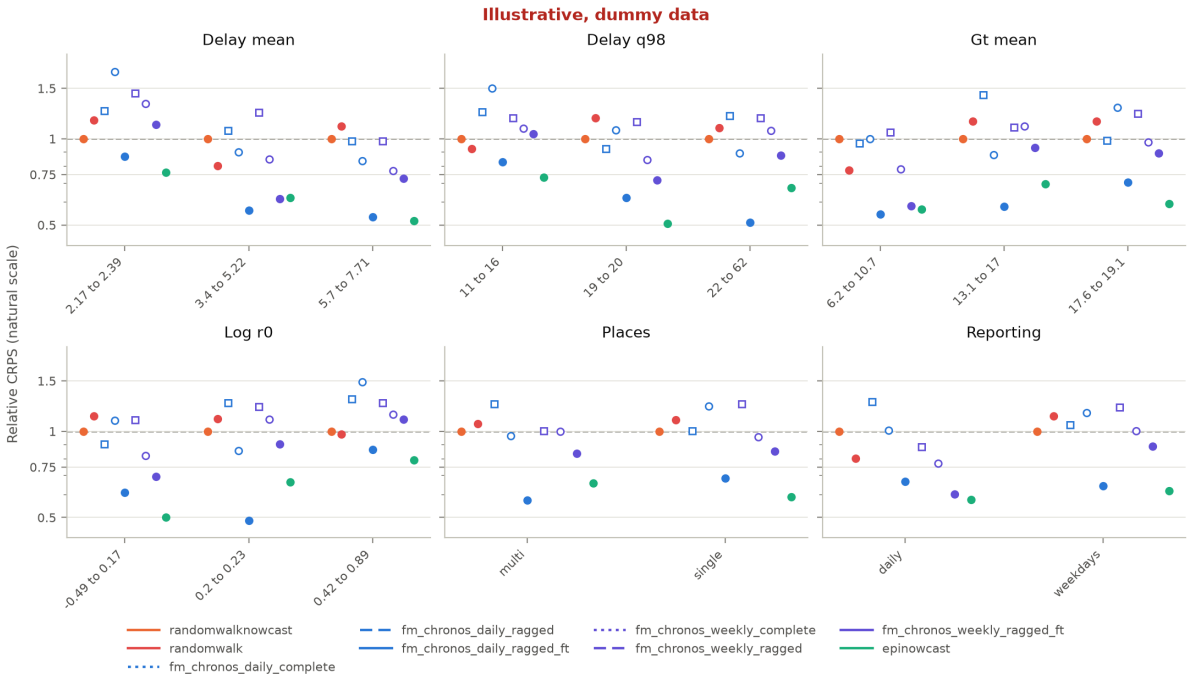


Figure 2: Held-out synthetic data: relative CRPS by level of each scenario factor. Illustrative, dummy data

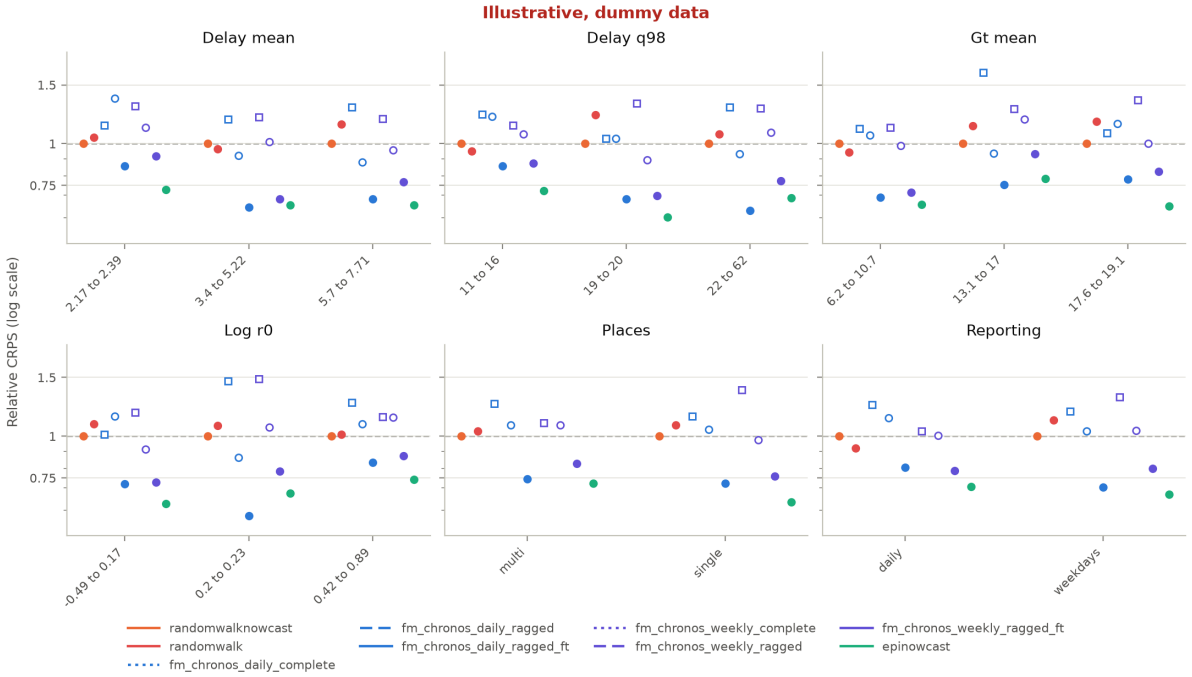


Figure 3: Held-out synthetic data: relative CRPS on the log scale by level of each scenario factor. Illustrative, dummy data

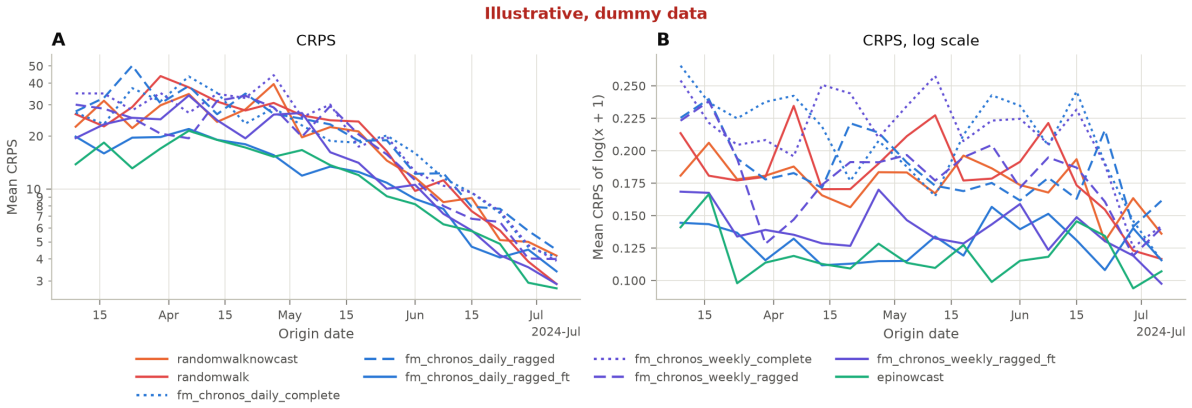


Figure 4: Held-out synthetic data: CRPS over time. Illustrative, dummy data

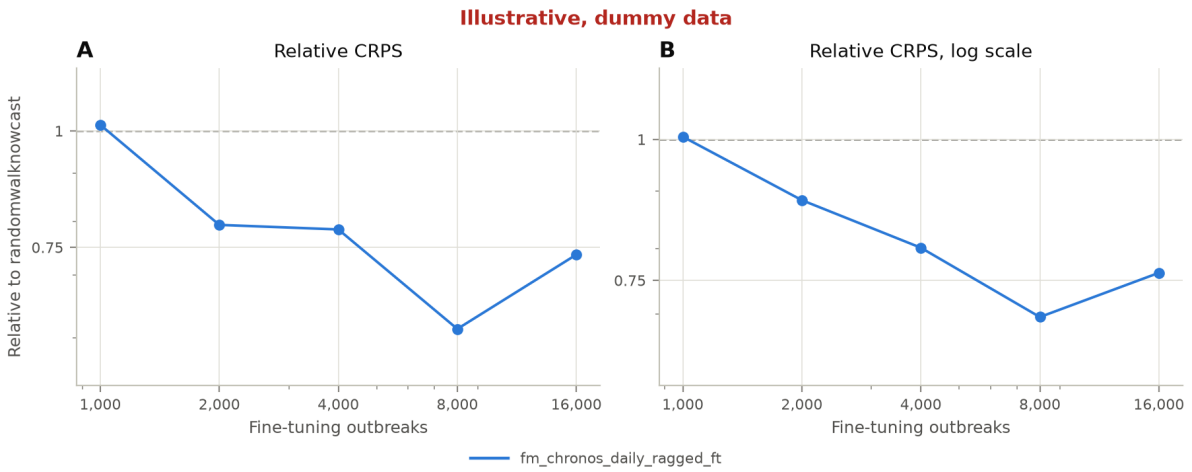


Figure 5: Held-out synthetic data: relative CRPS of each fine-tuning checkpoint, at doublings from 1,000 outbreaks, against the number of simulated outbreaks it was trained on (log scale). Illustrative, dummy data

Table 4: Held-out synthetic data scores by location and stratum. Illustrative, dummy data

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	epinow-crps	crps	s7001-sim01	all	20.9	0.1	0.0235	0.463	0.889	108	0.719	0.823
synthetic	epinow-crps	crps	s7001-sim02	all	9.25	0.135	0.0711	0.496	0.859	135	0.656	0.764
synthetic	epinow-crps	crps	s7001-sim03	all	18.8	0.101	0.12	0.481	0.933	135	0.382	0.484
synthetic	epinow-crps	crps	s7001-sim04	all	7.13	0.11	0.106	0.62	0.876	137	0.564	0.597
synthetic	epinow-crps	crps	s7001-sim05	all	18.9	0.107	0.399	0.455	0.818	132	0.91	0.765
synthetic	epinow-crps	crps	s7001-sim06	all	4.85	0.147	0.0372	0.664	0.905	137	0.732	0.705
synthetic	epinow-crps	crps	s7001-sim07	all	3.85	0.158	0.212	0.447	0.841	132	0.444	0.663
synthetic	epinow-crps	crps	s7001-sim08	all	18.8	0.0949	0.162	0.533	0.889	135	0.751	0.817
synthetic	fm_chronos_daily_ragged_ft	crps	s7001-sim01	all	38.3	0.209	-0.419	0.269	0.731	108	1.32	1.72

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	fm_chronos_daily_complete	pos	all	all	22.5	0.281	-0.13	0.437	0.83	135	1.6	1.59
synthetic	fm_chronos_daily_complete	pos	all	all	31.8	0.174	-0.331	0.393	0.741	135	0.646	0.832
synthetic	fm_chronos_daily_complete	pos	all	all	14.4	0.244	-0.288	0.401	0.803	137	1.14	1.33
synthetic	fm_chronos_daily_complete	pos	all	all	25.2	0.164	-0.152	0.402	0.871	132	1.21	1.17
synthetic	fm_chronos_daily_complete	pos	all	all	6.13	0.219	-0.265	0.445	0.854	137	0.925	1.05
synthetic	fm_chronos_daily_complete	pos	all	all	6.26	0.234	-0.0531	0.515	0.902	132	0.722	0.981
synthetic	fm_chronos_daily_complete	pos	all	all	36.9	0.164	-0.208	0.519	0.822	135	1.47	1.41
synthetic	fm_chronos_daily_ragged	pos	all	ragged	25.8	0.118	0.312	0.398	0.787	108	0.886	0.971
synthetic	fm_chronos_daily_ragged	pos	all	ragged	12.3	0.162	0.297	0.363	0.659	135	0.871	0.917
synthetic	fm_chronos_daily_ragged	pos	all	ragged	39.5	0.17	0.41	0.267	0.63	135	0.802	0.81
synthetic	fm_chronos_daily_ragged	pos	all	ragged	10.2	0.148	0.391	0.35	0.715	137	0.804	0.807
synthetic	fm_chronos_daily_ragged	pos	all	ragged	42.6	0.198	0.423	0.326	0.667	132	2.05	1.41
synthetic	fm_chronos_daily_ragged	pos	all	ragged	7.11	0.212	0.396	0.321	0.664	137	1.07	1.02
synthetic	fm_chronos_daily_ragged	pos	all	ragged	10.7	0.31	0.314	0.386	0.652	132	1.23	1.3
synthetic	fm_chronos_daily_ragged	pos	all	ragged	40.4	0.168	0.394	0.281	0.659	135	1.61	1.44
synthetic	fm_chronos_daily_ragged_ft	pos	all	ragged	13.6	0.0702	-0.0204	0.481	0.889	108	0.467	0.577
synthetic	fm_chronos_daily_ragged_ft	pos	all	ragged	10.8	0.151	0.0147	0.481	0.896	135	0.768	0.852
synthetic	fm_chronos_daily_ragged_ft	pos	all	ragged	24.4	0.123	-0.00748	0.511	0.904	135	0.496	0.589
synthetic	fm_chronos_daily_ragged_ft	pos	all	ragged	6.65	0.106	0.0326	0.569	0.912	137	0.526	0.579

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	fm_chronos_daily_ragged_ft	7001-sim05	ragged	21	0.142	0.351	0.439	0.795	132	1.01	1.02
synthetic	fm_chronos_daily_ragged_ft	7001-sim06	ragged	4.2	0.146	0.0411	0.533	0.898	137	0.634	0.702
synthetic	fm_chronos_daily_ragged_ft	7001-sim07	ragged	4.35	0.183	0.289	0.424	0.856	132	0.501	0.77
synthetic	fm_chronos_daily_ragged_ft	7001-sim08	ragged	21.9	0.0991	0.129	0.511	0.904	135	0.874	0.853
synthetic	fm_chronos_weekly_complete	7001-sim01	complete	33	0.191	-0.475	0.333	0.741	108	1.13	1.57
synthetic	fm_chronos_weekly_complete	7001-sim02	complete	14.5	0.196	-0.214	0.496	0.844	135	1.03	1.11
synthetic	fm_chronos_weekly_complete	7001-sim03	complete	43.9	0.225	-0.316	0.4	0.763	135	0.892	1.08
synthetic	fm_chronos_weekly_complete	7001-sim04	complete	17.4	0.265	-0.267	0.496	0.847	137	1.37	1.44
synthetic	fm_chronos_weekly_complete	7001-sim05	complete	31.5	0.201	-0.149	0.417	0.833	132	1.51	1.43
synthetic	fm_chronos_weekly_complete	7001-sim06	complete	6.4	0.206	-0.265	0.453	0.847	137	0.966	0.989
synthetic	fm_chronos_weekly_complete	7001-sim07	complete	5.49	0.234	-0.0997	0.515	0.886	132	0.633	0.981
synthetic	fm_chronos_weekly_complete	7001-sim08	complete	41.5	0.205	-0.413	0.281	0.763	135	1.65	1.76
synthetic	fm_chronos_weekly_ragged	7001-sim01	ragged	36.4	0.165	-0.147	0.333	0.769	108	1.25	1.35
synthetic	fm_chronos_weekly_ragged	7001-sim02	ragged	12.1	0.193	-0.0946	0.437	0.763	135	0.86	1.09
synthetic	fm_chronos_weekly_ragged	7001-sim03	ragged	25.7	0.139	-0.0736	0.407	0.778	135	0.523	0.663
synthetic	fm_chronos_weekly_ragged	7001-sim04	ragged	10.5	0.168	-0.112	0.504	0.869	137	0.828	0.914
synthetic	fm_chronos_weekly_ragged	7001-sim05	ragged	29.8	0.176	0.159	0.424	0.803	132	1.43	1.25
synthetic	fm_chronos_weekly_ragged	7001-sim06	ragged	5.9	0.229	-0.0977	0.467	0.781	137	0.891	1.1
synthetic	fm_chronos_weekly_ragged	7001-sim07	ragged	5.46	0.224	0.104	0.447	0.788	132	0.629	0.94

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	fm_chronos_weekly_ragged	crps	s7001-sim08	all	37.1	0.152	-0.0439	0.437	0.785	135	1.48	1.31
synthetic	fm_chronos_weekly_ragged_3	crps	s7001-sim01	all	11.8	0.14	0.0397	0.583	0.87	108	1.09	1.15
synthetic	fm_chronos_weekly_ragged_9	crps	s7001-sim02	all	9.35	0.144	0.146	0.585	0.941	135	0.664	0.813
synthetic	fm_chronos_weekly_ragged_28	crps	s7001-sim03	all	28.2	0.115	0.243	0.57	0.919	135	0.573	0.55
synthetic	fm_chronos_weekly_ragged_6	crps	s7001-sim04	all	6.98	0.109	0.0804	0.591	0.905	137	0.553	0.594
synthetic	fm_chronos_weekly_ragged_31	crps	s7001-sim05	all	31.8	0.158	0.458	0.432	0.78	132	1.53	1.13
synthetic	fm_chronos_weekly_ragged_4	crps	s7001-sim06	all	4.92	0.158	0.108	0.562	0.898	137	0.743	0.761
synthetic	fm_chronos_weekly_ragged_4	crps	s7001-sim07	all	4.35	0.183	0.244	0.508	0.856	132	0.501	0.769
synthetic	fm_chronos_weekly_ragged_25	crps	s7001-sim08	all	25.2	0.112	0.237	0.474	0.881	135	1.01	0.965
synthetic	random_walk	crps	s7001-sim01	all	38.7	0.181	-0.27	0.472	0.944	108	1.33	1.49
synthetic	random_walk	crps	s7001-sim02	all	12.1	0.166	-0.121	0.667	0.963	135	0.859	0.936
synthetic	random_walk	crps	s7001-sim03	all	52.6	0.244	-0.168	0.578	0.948	135	1.07	1.17
synthetic	random_walk	crps	s7001-sim04	all	8.82	0.158	-0.292	0.584	0.956	137	0.699	0.862
synthetic	random_walk	crps	s7001-sim05	all	21.9	0.146	-0.0563	0.553	0.955	132	1.05	1.04
synthetic	random_walk	crps	s7001-sim06	all	6.64	0.22	-0.218	0.555	0.934	137	1	1.06
synthetic	random_walk	crps	s7001-sim07	all	6.2	0.216	-0.0047	0.621	0.932	132	0.715	0.908
synthetic	random_walk	crps	s7001-sim08	all	35.2	0.153	-0.188	0.541	0.963	135	1.4	1.32
synthetic	random_walk_know-cast	crps	s7001-sim01	all	29.1	0.122	-0.00463	0.611	0.963	108	1	1

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	random_walk_nowcast	crps	s7001-sim02	all	14.1	0.177	0.0361	0.644	0.985	135	1	1
synthetic	random_walk_nowcast	crps	s7001-sim03	all	49.3	0.209	0.0698	0.615	0.97	135	1	1
synthetic	random_walk_nowcast	crps	s7001-sim04	all	12.6	0.184	0.0123	0.657	0.971	137	1	1
synthetic	random_walk_nowcast	crps	s7001-sim05	all	20.8	0.14	0.214	0.697	0.962	132	1	1
synthetic	random_walk_nowcast	crps	s7001-sim06	all	6.62	0.208	0.0144	0.679	0.971	137	1	1
synthetic	random_walk_nowcast	crps	s7001-sim07	all	8.67	0.238	0.24	0.629	0.962	132	1	1
synthetic	random_walk_nowcast	crps	s7001-sim08	all	25.1	0.116	0.0813	0.704	0.948	135	1	1

German nowcast hub

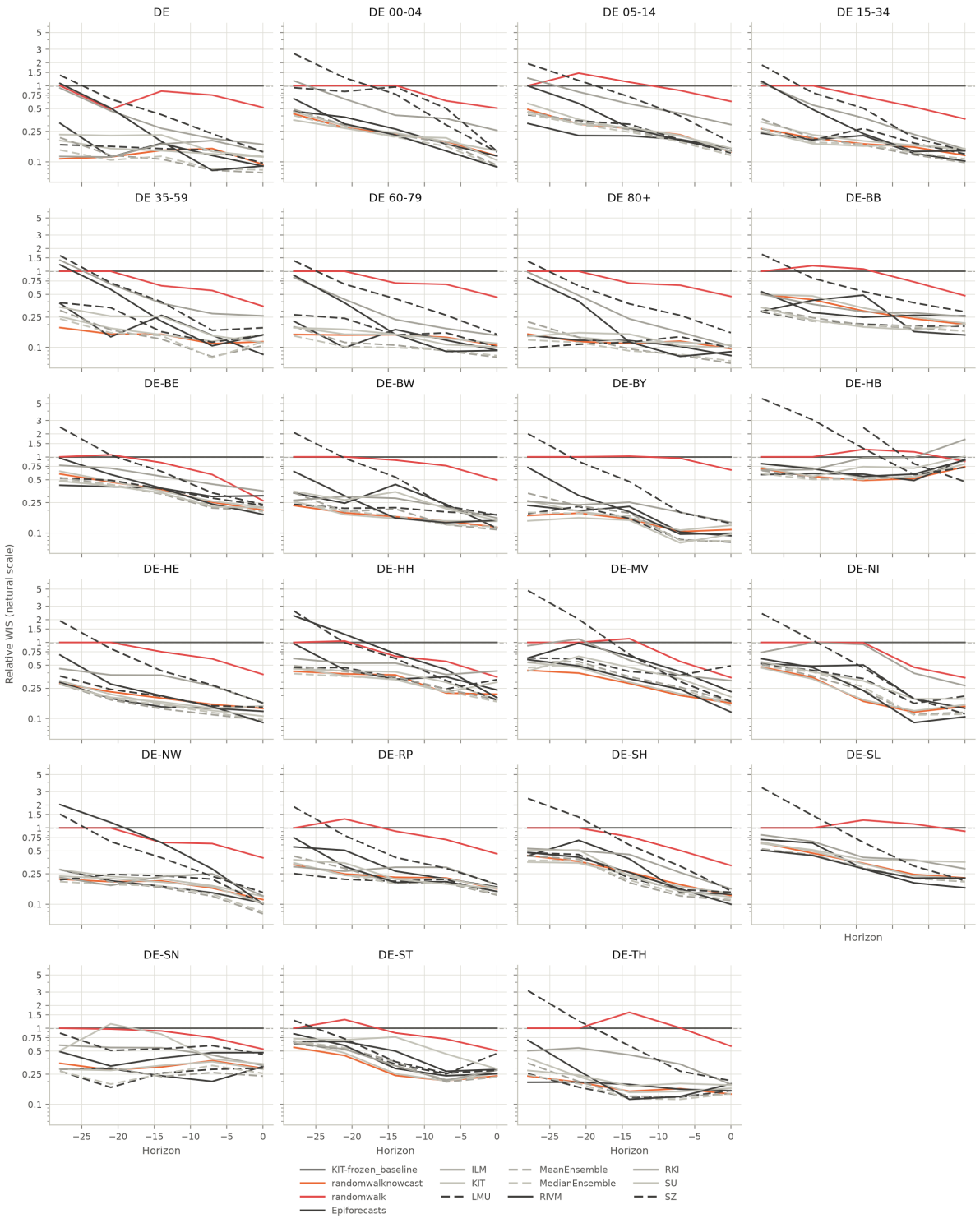


Figure 6: German nowcast hub: relative WIS by location and horizon.

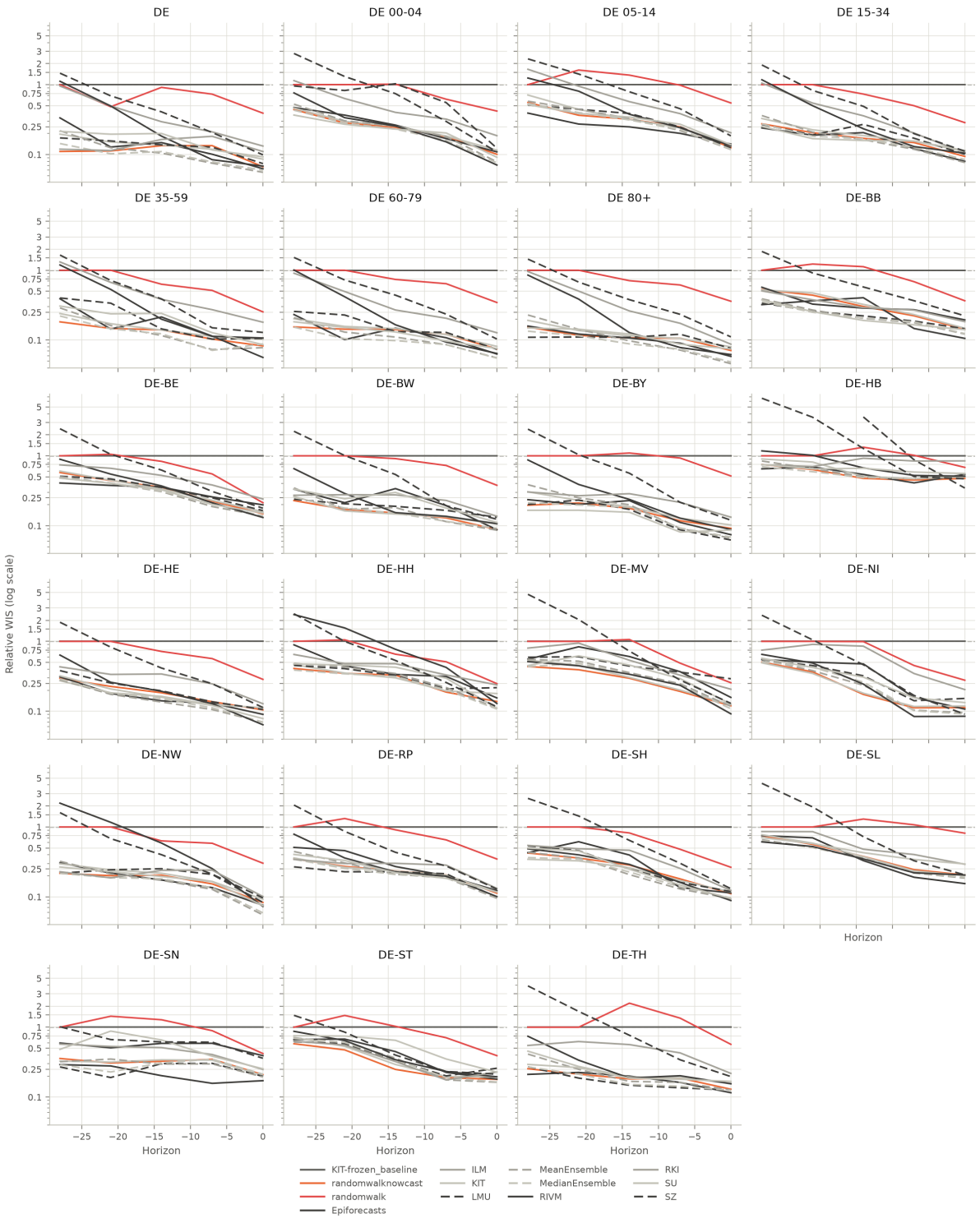


Figure 7: German nowcast hub: relative WIS on the log scale by location and horizon.

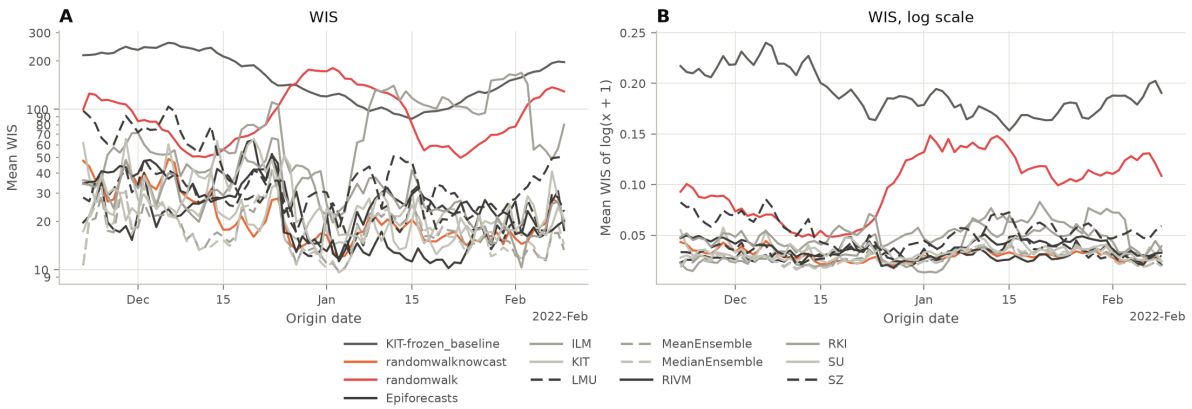


Figure 8: German nowcast hub: WIS over time, with a dotted line at the split.

Table 5: German nowcast hub scores by horizon.

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Epiforecasts	wis	-28	5.11	0.00657	-0.192	0.444	0.774	1837	0.302	0.411
germany	Epiforecasts	wis	-21	6.96	0.0117	0.126	0.378	0.763	1837	0.186	0.339
germany	Epiforecasts	wis	-14	17.8	0.0218	0.232	0.306	0.701	1837	0.217	0.281
germany	Epiforecasts	wis	-7	28.1	0.0344	0.0579	0.336	0.762	1837	0.144	0.177
germany	Epiforecasts	wis	0	55	0.0764	-0.0418	0.413	0.872	1837	0.111	0.121
germany	ILM	wis	-28	37.4	0.0194	0.783	0.214	0.61	518	0.98	1.15
germany	ILM	wis	-21	40.5	0.0218	0.717	0.282	0.689	560	0.494	0.61
germany	ILM	wis	-14	50.9	0.0277	0.662	0.305	0.725	560	0.284	0.358
germany	ILM	wis	-7	88.2	0.0464	0.615	0.334	0.605	560	0.207	0.245

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	ILM	wis	0	178	0.0891	0.611	0.261	0.546	518	0.17	0.143
germany	KIT	wis	-28	4.32	0.00684	0.125	0.608	0.935	1840	0.256	0.428
germany	KIT	wis	-21	7.31	0.0111	0.0219	0.566	0.932	1840	0.195	0.32
germany	KIT	wis	-14	14	0.0191	-0.0356	0.497	0.929	1840	0.171	0.246
germany	KIT	wis	-7	29	0.0366	-0.0226	0.467	0.929	1840	0.149	0.188
germany	KIT	wis	0	67.9	0.0865	-0.0386	0.493	0.917	1840	0.137	0.137
germany	KIT-frozen_baseline	wis	-28	16.9	0.016	-0.843	0.114	0.114	1840	1	1
germany	KIT-frozen_baseline	wis	-21	37.4	0.0347	-0.92	0.0538	0.0538	1840	1	1
germany	KIT-frozen_baseline	wis	-14	81.9	0.0778	-0.974	0.0185	0.0185	1840	1	1
germany	KIT-frozen_baseline	wis	-7	194	0.195	-0.99	0	0	1840	1	1
germany	KIT-frozen_baseline	wis	0	496	0.63	-0.979	0	0	1840	1	1
germany	LMU	wis	-28	4.36	0.00637	-0.163	0.41	0.77	1699	0.239	0.388
germany	LMU	wis	-21	8.73	0.0117	-0.108	0.292	0.672	1699	0.216	0.327
germany	LMU	wis	-14	16	0.0218	0.184	0.232	0.557	1699	0.181	0.273
germany	LMU	wis	-7	33.7	0.0396	0.0559	0.193	0.517	1699	0.16	0.198
germany	LMU	wis	0	69.2	0.0898	0.0541	0.217	0.536	1699	0.129	0.137
germany	MeanEnsemble	wis	-28	4.61	0.00682	0.203	0.473	0.834	1840	0.273	0.426

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	MeanEnsemble	wis	-21	6.81	0.0108	0.203	0.436	0.832	1840	0.182	0.312
germany	MeanEnsemble	wis	-14	12	0.018	0.205	0.39	0.796	1840	0.146	0.232
germany	MeanEnsemble	wis	-7	20.8	0.0317	0.139	0.389	0.819	1840	0.107	0.163
germany	MeanEnsemble	wis	0	46.4	0.0687	0.119	0.406	0.862	1817	0.0942	0.109
germany	MedianEnsemble	wis	-28	3.48	0.00576	0.0753	0.546	0.907	1840	0.206	0.36
germany	MedianEnsemble	wis	-21	5.91	0.0097	0.117	0.454	0.864	1840	0.158	0.279
germany	MedianEnsemble	wis	-14	12.1	0.0177	0.165	0.372	0.782	1840	0.148	0.227
germany	MedianEnsemble	wis	-7	21.8	0.0327	0.11	0.351	0.776	1840	0.112	0.168
germany	MedianEnsemble	wis	0	49.1	0.0703	0.081	0.387	0.82	1817	0.0997	0.112
germany	RIVM	wis	-28	16.4	0.0154	0.672	0.243	0.515	1840	0.972	0.963
germany	RIVM	wis	-21	18.2	0.0193	0.579	0.229	0.532	1840	0.487	0.557
germany	RIVM	wis	-14	18.7	0.026	0.392	0.224	0.546	1840	0.228	0.334
germany	RIVM	wis	-7	26	0.043	0.242	0.243	0.574	1840	0.134	0.221

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	RIVM	wis	0	60.5	0.0843	0.103	0.225	0.599	1840	0.122	0.134
germany	RKI	wis	-28	4.19	0.00819	-0.0231	0.288	0.441	1360	0.275	0.52
germany	RKI	wis	-21	8.6	0.016	0.192	0.208	0.339	1360	0.255	0.465
germany	RKI	wis	-14	20	0.0322	0.252	0.117	0.229	1360	0.27	0.411
germany	RKI	wis	-7	41.7	0.0611	0.323	0.0757	0.201	1360	0.238	0.309
germany	RKI	wis	0	76	0.127	0.331	0.128	0.307	1360	0.17	0.201
germany	SU	wis	-28	4.57	0.00659	-0.0846	0.473	0.833	1840	0.271	0.412
germany	SU	wis	-21	10.8	0.0134	0.0714	0.38	0.782	1840	0.288	0.385
germany	SU	wis	-14	20.9	0.0244	0.106	0.337	0.72	1840	0.255	0.313
germany	SU	wis	-7	30.7	0.0405	0.114	0.326	0.755	1840	0.158	0.208
germany	SU	wis	0	64.7	0.0855	0.191	0.365	0.813	1840	0.131	0.136
germany	SZ	wis	-28	26.4	0.0349	-0.276	0.0311	0.119	1771	1.54	2.17
germany	SZ	wis	-21	27.3	0.0357	-0.264	0.0924	0.252	1840	0.729	1.03
germany	SZ	wis	-14	36.5	0.0429	-0.249	0.146	0.368	1840	0.446	0.552
germany	SZ	wis	-7	49.4	0.0563	-0.152	0.207	0.512	1840	0.254	0.289
germany	SZ	wis	0	80.9	0.0985	-0.0429	0.264	0.588	1840	0.163	0.156
germany	randomwalk	wis	-28	16.9	0.016	-0.843	0.114	0.114	1840	1	1
germany	randomwalk	wis	-21	31.7	0.0387	-0.773	0.11	0.164	1840	0.847	1.11
germany	randomwalk	wis	-14	67.2	0.076	-0.255	0.291	0.641	1840	0.821	0.977
germany	randomwalk	wis	-7	138	0.135	-0.129	0.297	0.739	1840	0.709	0.695

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	randomwalk	wis	0	237	0.232	-0.0996	0.25	0.679	1840	0.477	0.368
germany	randomwalknowcast	wis	-28	3.37	0.00605	-0.226	0.636	0.917	1840	0.2	0.378
germany	randomwalknowcast	wis	-21	6.56	0.0103	-0.162	0.534	0.892	1840	0.175	0.297
germany	randomwalknowcast	wis	-14	13.9	0.0188	-0.131	0.458	0.882	1840	0.17	0.241
germany	randomwalknowcast	wis	-7	30.3	0.0362	-0.067	0.441	0.866	1840	0.156	0.186
germany	randomwalknowcast	wis	0	59	0.0786	-0.0973	0.559	0.946	1840	0.119	0.125

Table 6: German nowcast hub scores by location and stratum.

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Epiforecasts	wis	DE	00-04	3.37	0.0199	-0.366	0.422	0.902	400	0.155	0.149
germany	Epiforecasts	wis	DE	05-14	4.74	0.0282	0.0209	0.44	0.853	400	0.175	0.178
germany	Epiforecasts	wis	DE	15-34	14.8	0.0184	-0.00588	0.37	0.835	400	0.123	0.104

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Epiforecasts	wis	DE	35-59	40.5	0.0202	-0.188	0.233	0.623	400	0.121	0.087
germany	Epiforecasts	wis	DE	60-79	47.6	0.0182	-0.179	0.28	0.69	400	0.111	0.0836
germany	Epiforecasts	wis	DE	80+	31	0.0147	-0.14	0.307	0.795	400	0.0928	0.0735
germany	Epiforecasts	wis	DE	all	140	0.0174	-0.323	0.215	0.527	400	0.11	0.0842
germany	Epiforecasts	wis	DE-BB	all	9.79	0.038	0.388	0.355	0.777	400	0.208	0.159
germany	Epiforecasts	wis	DE-BE	all	9.62	0.0378	0.134	0.403	0.792	400	0.323	0.236
germany	Epiforecasts	wis	DE-BW	all	30.3	0.0265	-0.0144	0.318	0.693	400	0.18	0.132
germany	Epiforecasts	wis	DE-BY	all	27	0.02	-0.0891	0.297	0.713	400	0.116	0.0995
germany	Epiforecasts	wis	DE-HB	all	4.99	0.0538	0.208	0.608	0.87	400	0.786	0.522
germany	Epiforecasts	wis	DE-HE	all	9.66	0.0201	-0.0975	0.42	0.9	400	0.133	0.108
germany	Epiforecasts	wis	DE-HH	all	9.23	0.0566	-0.121	0.478	0.826	385	0.288	0.219
germany	Epiforecasts	wis	DE-MV	all	10.1	0.0438	0.349	0.33	0.735	400	0.324	0.249
germany	Epiforecasts	wis	DE-NI	all	8.45	0.0241	0.25	0.398	0.78	400	0.183	0.152

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Epiforecasts	wis	DE-NW	all	30.6	0.0225	-0.139	0.31	0.67	400	0.125	0.103
germany	Epiforecasts	wis	DE-RP	all	13	0.0404	0.161	0.32	0.718	400	0.211	0.17
germany	Epiforecasts	wis	DE-SH	all	4.82	0.0265	0.301	0.435	0.86	400	0.181	0.162
germany	Epiforecasts	wis	DE-SL	all	1.23	0.0228	0.195	0.667	0.98	400	0.217	0.217
germany	Epiforecasts	wis	DE-SN	all	37.2	0.0503	0.153	0.305	0.757	400	0.27	0.181
germany	Epiforecasts	wis	DE-ST	all	21.3	0.0548	0.00125	0.335	0.64	400	0.344	0.253
germany	Epiforecasts	wis	DE-TH	all	9.99	0.0202	0.333	0.39	0.875	400	0.158	0.134
germany	ILM	wis	DE	00-04	6.95	0.035	0.363	0.402	0.714	388	0.321	0.268
germany	ILM	wis	DE	05-14	10.9	0.0551	0.647	0.157	0.474	388	0.413	0.358
germany	ILM	wis	DE	15-34	25.3	0.0303	0.696	0.348	0.755	388	0.215	0.177
germany	ILM	wis	DE	35-59	101	0.0554	0.822	0.191	0.546	388	0.313	0.247
germany	ILM	wis	DE	60-79	79.2	0.039	0.74	0.278	0.686	388	0.193	0.184
germany	ILM	wis	DE	80+	56.1	0.0301	0.653	0.369	0.698	388	0.175	0.154
germany	ILM	wis	DE	all	267	0.0383	0.818	0.219	0.585	388	0.219	0.191
germany	KIT	wis	DE	00-04	3.6	0.0203	-0.119	0.583	0.948	400	0.165	0.152
germany	KIT	wis	DE	05-14	5.35	0.0329	0.118	0.477	0.902	400	0.198	0.208

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	KIT	wis	DE	15-34	18.9	0.0234	-0.00188	0.537	0.988	400	0.157	0.132
germany	KIT	wis	DE	35-59	48	0.0259	0.138	0.557	0.99	400	0.143	0.111
germany	KIT	wis	DE	60-79	53.2	0.0217	0.0908	0.527	0.988	400	0.124	0.0994
germany	KIT	wis	DE	80+	38.1	0.0185	0.0751	0.59	0.988	400	0.114	0.0927
germany	KIT	wis	DE	all	157	0.0209	0.0929	0.56	0.993	400	0.124	0.101
germany	KIT	wis	DE-BB	all	12.1	0.0492	-0.151	0.588	0.912	400	0.256	0.206
germany	KIT	wis	DE-BE	all	7.86	0.0334	0.0665	0.463	0.91	400	0.263	0.209
germany	KIT	wis	DE-BW	all	24.8	0.0235	0.128	0.502	0.963	400	0.147	0.118
germany	KIT	wis	DE-BY	all	30.7	0.0243	0.266	0.455	0.875	400	0.131	0.121
germany	KIT	wis	DE-HB	all	4.58	0.0527	-0.0318	0.58	0.858	400	0.721	0.512
germany	KIT	wis	DE-HE	all	10.9	0.023	-0.244	0.525	0.93	400	0.15	0.123
germany	KIT	wis	DE-HH	all	9.57	0.0527	-0.0104	0.527	0.943	400	0.289	0.202
germany	KIT	wis	DE-MV	all	6.17	0.0287	-0.0261	0.502	0.93	400	0.199	0.163
germany	KIT	wis	DE-NI	all	7.28	0.0211	-0.0126	0.547	0.907	400	0.157	0.133
germany	KIT	wis	DE-NW	all	37.1	0.0271	-0.119	0.522	0.955	400	0.152	0.124
germany	KIT	wis	DE-RP	all	11.6	0.0369	0.0598	0.42	0.885	400	0.188	0.155
germany	KIT	wis	DE-SH	all	4.41	0.0247	-0.267	0.62	0.897	400	0.165	0.151
germany	KIT	wis	DE-SL	all	1.46	0.0269	-0.016	0.632	0.938	400	0.257	0.256
germany	KIT	wis	DE-SN	all	43.3	0.0717	0.0872	0.525	0.94	400	0.314	0.258

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	KIT	wis	DE-ST	all	16.8	0.051	-0.106	0.405	0.802	400	0.271	0.236
germany	KIT	wis	DE-TH	all	10.4	0.0261	0.212	0.458	0.917	400	0.164	0.172
germany	KIT-frozen_baseline	wis	DE	00-04	21.8	0.133	-0.912	0.0625	0.0625	400	1	1
germany	KIT-frozen_baseline	wis	DE	05-14	27.1	0.159	-0.875	0.035	0.035	400	1	1
germany	KIT-frozen_baseline	wis	DE	15-34	120	0.177	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE	35-59	335	0.232	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE	60-79	427	0.218	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE	80+	334	0.2	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE	all	1.27e+03	3.207	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE-BB	all	47.2	0.239	-0.988	0.0075	0.0075	400	1	1
germany	KIT-frozen_baseline	wis	DE-BE	all	29.8	0.16	-0.948	0.0425	0.0425	400	1	1
germany	KIT-frozen_baseline	wis	DE-BW	all	169	0.2	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE-BY	all	233	0.201	-1	0	0	400	1	1

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	KIT-frozen_baseline	wis	DE-HB	all	6.35	0.103	-0.585	0.245	0.245	400	1	1
germany	KIT-frozen_baseline	wis	DE-HE	all	72.6	0.187	-0.988	0.0125	0.0125	400	1	1
germany	KIT-frozen_baseline	wis	DE-HH	all	33.1	0.261	-0.965	0.035	0.035	400	1	1
germany	KIT-frozen_baseline	wis	DE-MV	all	31.1	0.176	-0.907	0.0675	0.0675	400	1	1
germany	KIT-frozen_baseline	wis	DE-NI	all	46.2	0.158	-0.853	0.0125	0.0125	400	1	1
germany	KIT-frozen_baseline	wis	DE-NW	all	244	0.218	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE-RP	all	61.7	0.238	-0.98	0.01	0.01	400	1	1
germany	KIT-frozen_baseline	wis	DE-SH	all	26.7	0.164	-0.955	0.045	0.045	400	1	1
germany	KIT-frozen_baseline	wis	DE-SL	all	5.66	0.105	-0.735	0.245	0.245	400	1	1
germany	KIT-frozen_baseline	wis	DE-SN	all	138	0.278	-1	0	0	400	1	1
germany	KIT-frozen_baseline	wis	DE-ST	all	62.1	0.217	-0.978	0.0225	0.0225	400	1	1
germany	KIT-frozen_baseline	wis	DE-TH	all	63.2	0.151	-0.985	0.015	0.015	400	1	1
germany	LMU	wis	DE	00-04	6.52	0.0412	0.783	0.16	0.5	400	0.299	0.308

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	LMU	wis	DE	05-14	4.88	0.0321	0.485	0.305	0.69	400	0.18	0.202
germany	LMU	wis	DE	15-34	18.5	0.0235	0.377	0.247	0.608	400	0.154	0.133
germany	LMU	wis	DE	35-59	50.6	0.0277	-0.452	0.128	0.34	400	0.151	0.119
germany	LMU	wis	DE	60-79	55.4	0.0215	-0.494	0.122	0.385	400	0.13	0.0987
germany	LMU	wis	DE	80+	36.7	0.018	-0.145	0.245	0.605	400	0.11	0.0903
germany	LMU	wis	DE	all	148	0.0196	-0.303	0.102	0.302	400	0.117	0.0948
germany	LMU	wis	DE-BB	all	9.23	0.0409	-0.0773	0.338	0.745	400	0.196	0.171
germany	LMU	wis	DE-BE	all	8.27	0.0348	0.108	0.292	0.67	400	0.277	0.217
germany	LMU	wis	DE-BW	all	31	0.0293	-0.17	0.282	0.532	400	0.184	0.146
germany	LMU	wis	DE-BY	all	22.1	0.0167	0.0131	0.255	0.605	400	0.0947	0.0828
germany	LMU	wis	DE-HB	all	2.42	0.0483	0.394	0.589	0.874	95	0.61	0.54
germany	LMU	wis	DE-HE	all	11.2	0.0238	-0.189	0.333	0.705	400	0.154	0.127
germany	LMU	wis	DE-HH	all	10.4	0.0611	-0.218	0.345	0.725	400	0.315	0.234
germany	LMU	wis	DE-MV	all	14.4	0.0574	0.0636	0.305	0.64	400	0.464	0.326
germany	LMU	wis	DE-NI	all	9.64	0.0273	0.16	0.305	0.735	400	0.208	0.172
germany	LMU	wis	DE-NW	all	35.5	0.0279	-0.186	0.182	0.487	400	0.145	0.128
germany	LMU	wis	DE-RP	all	10.1	0.0336	0.0461	0.355	0.757	400	0.164	0.141
germany	LMU	wis	DE-SH	all	4.47	0.0256	-0.146	0.505	0.882	400	0.167	0.156
germany	LMU	wis	DE-SN	all	38.5	0.0653	0.25	0.175	0.445	400	0.279	0.235

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	LMU	wis	DE-ST	all	25.4	0.0606	-0.0419	0.195	0.557	400	0.409	0.28
germany	LMU	wis	DE-TH	all	9.17	0.02	0.138	0.39	0.84	400	0.145	0.132
germany	MeanEnsemble	wis	DE	00-04	2.83	0.017	0.218	0.496	0.917	399	0.13	0.128
germany	MeanEnsemble	wis	DE	05-14	4.52	0.0298	0.334	0.346	0.815	399	0.168	0.189
germany	MeanEnsemble	wis	DE	15-34	14	0.0177	0.352	0.416	0.855	399	0.117	0.101
germany	MeanEnsemble	wis	DE	35-59	35.2	0.0196	0.392	0.378	0.83	399	0.106	0.0852
germany	MeanEnsemble	wis	DE	60-79	36.6	0.0156	0.236	0.368	0.862	399	0.0863	0.0719
germany	MeanEnsemble	wis	DE	80+	25.5	0.0125	0.246	0.416	0.897	399	0.0771	0.063
germany	MeanEnsemble	wis	DE	all	103	0.0142	0.315	0.361	0.807	399	0.0822	0.0689
germany	MeanEnsemble	wis	DE-BB	all	8.47	0.0368	0.0677	0.434	0.87	399	0.181	0.155
germany	MeanEnsemble	wis	DE-BE	all	6.86	0.0287	0.145	0.411	0.807	399	0.232	0.181
germany	MeanEnsemble	wis	DE-BW	all	21.7	0.0207	0.131	0.333	0.767	399	0.13	0.104
germany	MeanEnsemble	wis	DE-BY	all	23.5	0.0184	0.201	0.401	0.729	399	0.102	0.0919

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	MeanEnsemble	wis	DE-HB	all	4.88	0.0528	0.125	0.594	0.877	399	0.771	0.515
germany	MeanEnsemble	wis	DE-HE	all	7.77	0.0164	-0.144	0.494	0.917	399	0.108	0.088
germany	MeanEnsemble	wis	DE-HH	all	7.33	0.043	0.154	0.411	0.842	399	0.222	0.166
germany	MeanEnsemble	wis	DE-MV	all	6.68	0.0302	0.0893	0.398	0.825	399	0.217	0.173
germany	MeanEnsemble	wis	DE-NI	all	6.35	0.0186	0.114	0.429	0.88	399	0.138	0.118
germany	MeanEnsemble	wis	DE-NW	all	25.1	0.019	0.156	0.358	0.769	399	0.103	0.0875
germany	MeanEnsemble	wis	DE-RP	all	10	0.0325	0.123	0.348	0.784	399	0.163	0.137
germany	MeanEnsemble	wis	DE-SH	all	3.62	0.0204	-0.0353	0.566	0.92	399	0.136	0.125
germany	MeanEnsemble	wis	DE-SL	all	1.31	0.0247	0.126	0.664	0.972	399	0.232	0.236
germany	MeanEnsemble	wis	DE-SN	all	34.1	0.0663	0.379	0.293	0.604	399	0.25	0.241
germany	MeanEnsemble	wis	DE-ST	all	16.8	0.0478	0.0183	0.261	0.657	399	0.271	0.221
germany	MeanEnsemble	wis	DE-TH	all	8.78	0.0214	0.257	0.449	0.85	399	0.14	0.142

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	MediaNEnsemble	wis	DE	00-04	2.63	0.0156	0.0627	0.511	0.95	399	0.121	0.118
germany	MediaNEnsemble	wis	DE	05-14	4.42	0.0294	0.292	0.398	0.794	399	0.164	0.187
germany	MediaNEnsemble	wis	DE	15-34	14.9	0.0186	0.312	0.371	0.83	399	0.125	0.106
germany	MediaNEnsemble	wis	DE	35-59	37.6	0.0209	0.233	0.401	0.845	399	0.113	0.0907
germany	MediaNEnsemble	wis	DE	60-79	36.6	0.0149	0.126	0.396	0.867	399	0.0864	0.0687
germany	MediaNEnsemble	wis	DE	80+	25.3	0.0122	0.184	0.409	0.887	399	0.0765	0.0616
germany	MediaNEnsemble	wis	DE	all	108	0.0145	0.193	0.366	0.802	399	0.0857	0.0706
germany	MediaNEnsemble	wis	DE-BB	all	8.45	0.0369	-0.0143	0.434	0.83	399	0.18	0.156
germany	MediaNEnsemble	wis	DE-BE	all	6.89	0.0292	0.0917	0.404	0.812	399	0.233	0.184

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	MediaNEnsemble	wis	DE-BW	all	21.9	0.0209	0.0554	0.358	0.754	399	0.131	0.105
germany	MediaNEnsemble	wis	DE-BY	all	22	0.0177	0.224	0.404	0.799	399	0.0956	0.0884
germany	MediaNEnsemble	wis	DE-HB	all	4.88	0.0528	0.125	0.594	0.872	399	0.769	0.515
germany	MediaNEnsemble	wis	DE-HE	all	8.05	0.0169	-0.156	0.449	0.902	399	0.112	0.091
germany	MediaNEnsemble	wis	DE-HH	all	6.85	0.04	0.0377	0.481	0.885	399	0.207	0.154
germany	MediaNEnsemble	wis	DE-MV	all	6.09	0.0282	0.103	0.388	0.81	399	0.198	0.162
germany	MediaNEnsemble	wis	DE-NI	all	6.23	0.0184	0.109	0.404	0.865	399	0.136	0.117
germany	MediaNEnsemble	wis	DE-NW	all	25.9	0.0195	-0.0753	0.383	0.789	399	0.106	0.0898
germany	MediaNEnsemble	wis	DE-RP	all	10	0.0326	0.0856	0.386	0.799	399	0.163	0.137

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	MediaNEnsemble	wis	DE-SH	all	3.73	0.0209	-0.122	0.561	0.907	399	0.14	0.129
germany	MediaNEnsemble	wis	DE-SL	all	1.32	0.0249	0.0952	0.624	0.97	399	0.233	0.238
germany	MediaNEnsemble	wis	DE-SN	all	36.7	0.0707	0.3	0.268	0.564	399	0.269	0.257
germany	MediaNEnsemble	wis	DE-ST	all	16.7	0.0482	0.0119	0.261	0.642	399	0.27	0.223
germany	MediaNEnsemble	wis	DE-TH	all	8.44	0.0201	0.25	0.459	0.907	399	0.135	0.134
germany	RIVM	wis	DE	00-04	2.68	0.0165	0.29	0.453	0.88	400	0.123	0.124
germany	RIVM	wis	DE	05-14	5.53	0.0384	0.472	0.242	0.588	400	0.205	0.242
germany	RIVM	wis	DE	15-34	21	0.0265	0.658	0.152	0.417	400	0.175	0.15
germany	RIVM	wis	DE	35-59	59.5	0.033	0.751	0.065	0.27	400	0.177	0.142
germany	RIVM	wis	DE	60-79	54.9	0.0233	0.584	0.122	0.33	400	0.129	0.107
germany	RIVM	wis	DE	80+	40.8	0.0196	0.51	0.102	0.37	400	0.122	0.0981
germany	RIVM	wis	DE	all	170	0.0234	0.674	0.0925	0.235	400	0.134	0.113
germany	RIVM	wis	DE-BB	all	12.7	0.0561	0.219	0.22	0.537	400	0.268	0.235
germany	RIVM	wis	DE-BE	all	7.39	0.0319	0.421	0.21	0.575	400	0.248	0.199

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	RIVM	wis	DE-BW	all	26.9	0.0256	0.528	0.185	0.507	400	0.159	0.128
germany	RIVM	wis	DE-BY	all	30.6	0.0258	0.738	0.133	0.445	400	0.131	0.128
germany	RIVM	wis	DE-HB	all	5.07	0.0567	0.363	0.495	0.815	400	0.798	0.551
germany	RIVM	wis	DE-HE	all	9.05	0.0187	0.163	0.3	0.71	400	0.125	0.1
germany	RIVM	wis	DE-HH	all	12	0.0819	0.501	0.147	0.403	400	0.362	0.313
germany	RIVM	wis	DE-MV	all	5.55	0.0262	0.106	0.333	0.752	400	0.179	0.149
germany	RIVM	wis	DE-NI	all	5.85	0.0176	0.129	0.347	0.805	400	0.126	0.111
germany	RIVM	wis	DE-NW	all	65	0.0466	0.578	0.107	0.3	400	0.266	0.213
germany	RIVM	wis	DE-RP	all	11.1	0.0359	0.356	0.19	0.62	400	0.18	0.151
germany	RIVM	wis	DE-SH	all	3.82	0.0224	-0.0484	0.383	0.828	400	0.143	0.137
germany	RIVM	wis	DE-SL	all	1.38	0.0258	0.152	0.593	0.94	400	0.244	0.246
germany	RIVM	wis	DE-SN	all	62.8	0.13	0.347	0.09	0.273	400	0.456	0.469
germany	RIVM	wis	DE-ST	all	18.3	0.0558	0.198	0.142	0.435	400	0.294	0.258
germany	RIVM	wis	DE-TH	all	11.3	0.0268	0.45	0.253	0.688	400	0.178	0.177
germany	RKI	wis	DE	all	191	0.0268	0.076	0.11	0.343	400	0.151	0.13
germany	RKI	wis	DE-BB	all	13.2	0.0547	-0.217	0.125	0.273	400	0.28	0.228
germany	RKI	wis	DE-BE	all	12.5	0.0514	0.333	0.135	0.235	400	0.421	0.321
germany	RKI	wis	DE-BW	all	32	0.0341	0.409	0.18	0.355	400	0.19	0.17
germany	RKI	wis	DE-BY	all	39.3	0.0337	0.217	0.177	0.32	400	0.168	0.167

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	RKI	wis	DE-HB	all	9.03	0.087	0.507	0.343	0.425	400	1.42	0.845
germany	RKI	wis	DE-HE	all	15.5	0.0333	0.303	0.158	0.275	400	0.213	0.178
germany	RKI	wis	DE-HH	all	14.2	0.0755	-0.154	0.147	0.27	400	0.429	0.289
germany	RKI	wis	DE-MV	all	11.6	0.0484	0.485	0.128	0.287	400	0.373	0.275
germany	RKI	wis	DE-NI	all	16.7	0.046	0.691	0.115	0.215	400	0.362	0.29
germany	RKI	wis	DE-NW	all	41.1	0.0313	0.00725	0.158	0.345	400	0.168	0.143
germany	RKI	wis	DE-RP	all	14.2	0.0449	-0.202	0.182	0.305	400	0.23	0.188
germany	RKI	wis	DE-SH	all	6	0.0323	0.291	0.225	0.38	400	0.225	0.197
germany	RKI	wis	DE-SL	all	2.01	0.0393	0.38	0.297	0.47	400	0.356	0.375
germany	RKI	wis	DE-SN	all	58	0.0935	-0.178	0.0625	0.128	400	0.421	0.336
germany	RKI	wis	DE-ST	all	18.3	0.0548	-0.0451	0.133	0.295	400	0.295	0.253
germany	RKI	wis	DE-TH	all	16.5	0.0447	0.752	0.102	0.24	400	0.262	0.295
germany	SU	wis	DE	00-04	3.07	0.0178	0.026	0.502	0.92	400	0.141	0.134
germany	SU	wis	DE	05-14	5.03	0.0331	0.24	0.357	0.767	400	0.186	0.209
germany	SU	wis	DE	15-34	18.8	0.023	0.204	0.34	0.755	400	0.156	0.13
germany	SU	wis	DE	35-59	48.6	0.0264	-0.0117	0.237	0.65	400	0.145	0.114
germany	SU	wis	DE	60-79	48.5	0.0195	0.0104	0.36	0.797	400	0.113	0.0894
germany	SU	wis	DE	80+	36.5	0.0171	0.0574	0.362	0.833	400	0.109	0.0855
germany	SU	wis	DE	all	177	0.0231	-0.115	0.207	0.61	400	0.14	0.112

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	SU	wis	DE-BB	all	9.27	0.0386	0.027	0.42	0.863	400	0.197	0.161
germany	SU	wis	DE-BE	all	7.3	0.0311	-0.174	0.422	0.83	400	0.245	0.194
germany	SU	wis	DE-BW	all	31.3	0.0288	0.0686	0.315	0.74	400	0.185	0.144
germany	SU	wis	DE-BY	all	23.4	0.019	-0.0814	0.398	0.85	400	0.1	0.0942
germany	SU	wis	DE-HB	all	5.74	0.0595	0.0369	0.53	0.807	400	0.904	0.578
germany	SU	wis	DE-HE	all	8.88	0.0188	-0.09	0.455	0.88	400	0.122	0.1
germany	SU	wis	DE-HH	all	7.97	0.0492	0.211	0.398	0.815	400	0.24	0.188
germany	SU	wis	DE-MV	all	8.06	0.0345	0.156	0.388	0.752	400	0.259	0.196
germany	SU	wis	DE-NI	all	9.27	0.0253	0.103	0.388	0.843	400	0.201	0.16
germany	SU	wis	DE-NW	all	32.5	0.0239	-0.0363	0.323	0.73	400	0.133	0.109
germany	SU	wis	DE-RP	all	11.3	0.0362	-0.0413	0.417	0.812	400	0.183	0.152
germany	SU	wis	DE-SH	all	4.19	0.0223	-0.0366	0.51	0.897	400	0.157	0.136
germany	SU	wis	DE-SL	all	2.13	0.0358	0.249	0.53	0.95	400	0.377	0.342
germany	SU	wis	DE-SN	all	68.8	0.102	0.568	0.212	0.485	400	0.499	0.367
germany	SU	wis	DE-ST	all	26.4	0.0722	0.136	0.19	0.49	400	0.426	0.334
germany	SU	wis	DE-TH	all	11.7	0.0265	0.323	0.393	0.875	400	0.185	0.175
germany	SZ	wis	DE	00-04	6.44	0.0398	-0.461	0.118	0.421	397	0.294	0.296
germany	SZ	wis	DE	05-14	10	0.066	-0.329	0.101	0.32	397	0.368	0.414
germany	SZ	wis	DE	15-34	28.3	0.0374	-0.339	0.134	0.325	397	0.233	0.209

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	SZ	wis	DE	35-59	81.5	0.0442	-0.13	0.126	0.297	397	0.241	0.189
germany	SZ	wis	DE	60-79	112	0.0447	-0.166	0.126	0.275	397	0.26	0.203
germany	SZ	wis	DE	80+	86.6	0.0411	-0.208	0.0907	0.264	397	0.257	0.205
germany	SZ	wis	DE	all	300	0.0393	-0.178	0.101	0.252	397	0.235	0.189
germany	SZ	wis	DE-BB	all	19.5	0.0848	-0.254	0.131	0.373	397	0.411	0.352
germany	SZ	wis	DE-BE	all	11.5	0.0513	-0.0358	0.209	0.453	397	0.383	0.318
germany	SZ	wis	DE-BW	all	46.8	0.0438	-0.129	0.154	0.31	397	0.276	0.217
germany	SZ	wis	DE-BY	all	59.9	0.0471	0.036	0.146	0.35	397	0.255	0.232
germany	SZ	wis	DE-HB	all	5.72	0.0752	-0.191	0.166	0.383	397	0.895	0.725
germany	SZ	wis	DE-HE	all	18.6	0.0393	-0.382	0.146	0.38	397	0.254	0.209
germany	SZ	wis	DE-HH	all	10.3	0.0616	-0.0742	0.202	0.451	397	0.309	0.234
germany	SZ	wis	DE-MV	all	10.2	0.0504	-0.188	0.234	0.524	397	0.327	0.284
germany	SZ	wis	DE-NI	all	9.67	0.0287	-0.23	0.189	0.531	397	0.208	0.18
germany	SZ	wis	DE-NW	all	56.4	0.0401	-0.328	0.134	0.36	397	0.229	0.182
germany	SZ	wis	DE-RP	all	17.7	0.0569	-0.151	0.159	0.365	397	0.285	0.237
germany	SZ	wis	DE-SH	all	8.11	0.0479	-0.338	0.186	0.416	397	0.302	0.29
germany	SZ	wis	DE-SL	all	2.31	0.0473	-0.0694	0.262	0.562	397	0.404	0.447
germany	SZ	wis	DE-SN	all	72.7	0.133	0.063	0.126	0.252	397	0.524	0.477
germany	SZ	wis	DE-ST	all	21.5	0.0655	-0.388	0.0982	0.325	397	0.344	0.3

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	SZ	wis	DE-TH	all	21.6	0.0519	-0.0399	0.0907	0.312	397	0.34	0.34
germany	random_walk	wis	DE	00-04	12.9	0.0712	-0.601	0.242	0.417	400	0.591	0.534
germany	random_walk	wis	DE	05-14	20.9	0.124	-0.354	0.172	0.417	400	0.773	0.784
germany	random_walk	wis	DE	15-34	55.6	0.0703	-0.634	0.175	0.427	400	0.461	0.397
germany	random_walk	wis	DE	35-59	155	0.0853	-0.469	0.163	0.482	400	0.463	0.367
germany	random_walk	wis	DE	60-79	245	0.103	-0.393	0.138	0.412	400	0.573	0.474
germany	random_walk	wis	DE	80+	192	0.0956	-0.379	0.172	0.512	400	0.576	0.479
germany	random_walk	wis	DE	all	778	0.107	-0.181	0.138	0.315	400	0.615	0.516
germany	random_walk	wis	DE-BB	all	31.3	0.134	-0.332	0.21	0.51	400	0.663	0.56
germany	random_walk	wis	DE-BE	all	13.4	0.0616	-0.435	0.362	0.555	400	0.45	0.384
germany	random_walk	wis	DE-BW	all	105	0.102	-0.434	0.152	0.42	400	0.62	0.51
germany	random_walk	wis	DE-BY	all	186	0.133	-0.358	0.145	0.41	400	0.799	0.663
germany	random_walk	wis	DE-HB	all	6.19	0.0839	-0.302	0.343	0.535	400	0.974	0.815
germany	random_walk	wis	DE-HE	all	35.6	0.0759	-0.545	0.258	0.492	400	0.49	0.405
germany	random_walk	wis	DE-HH	all	15.4	0.0959	-0.677	0.21	0.46	400	0.463	0.367
germany	random_walk	wis	DE-MV	all	14.8	0.0668	-0.361	0.34	0.542	400	0.476	0.379
germany	random_walk	wis	DE-NI	all	20.1	0.0596	-0.617	0.11	0.352	400	0.435	0.376
germany	random_walk	wis	DE-NW	all	124	0.0908	-0.563	0.158	0.458	400	0.509	0.416
germany	random_walk	wis	DE-RP	all	37.6	0.12	-0.451	0.242	0.53	400	0.61	0.506

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	random_walk	wis	DE-SH	all	11.6	0.064	-0.701	0.225	0.455	400	0.436	0.391
germany	random_walk	wis	DE-SL	all	5.72	0.0988	-0.26	0.393	0.63	400	1.01	0.943
germany	random_walk	wis	DE-SN	all	97	0.191	-0.0543	0.225	0.593	400	0.704	0.689
germany	random_walk	wis	DE-ST	all	41.3	0.128	-0.292	0.133	0.45	400	0.665	0.589
germany	random_walk	wis	DE-TH	all	50.3	0.127	-0.267	0.18	0.372	400	0.796	0.84
germany	random_walk_nowcast	wis	DE	00-04	3.28	0.0186	-0.264	0.56	0.938	400	0.15	0.14
germany	random_walk_nowcast	wis	DE	05-14	5.03	0.0307	-0.0954	0.475	0.835	400	0.186	0.194
germany	random_walk_nowcast	wis	DE	15-34	16.7	0.0207	-0.138	0.522	0.98	400	0.138	0.117
germany	random_walk_nowcast	wis	DE	35-59	41.1	0.0219	-0.0571	0.532	0.938	400	0.122	0.0944
germany	random_walk_nowcast	wis	DE	60-79	51.5	0.0204	-0.17	0.578	0.953	400	0.12	0.0936
germany	random_walk_nowcast	wis	DE	80+	35.4	0.0169	-0.224	0.603	0.955	400	0.106	0.0847
germany	random_walk_nowcast	wis	DE	all	142	0.0183	-0.0729	0.34	0.815	400	0.112	0.0885

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	randomwalknowcast	wis	DE-BB	all	11.4	0.0472	-0.3	0.545	0.875	400	0.242	0.197
germany	randomwalknowcast	wis	DE-BE	all	7.49	0.0323	-0.0339	0.497	0.892	400	0.251	0.202
germany	randomwalknowcast	wis	DE-BW	all	22.4	0.021	-0.0239	0.545	0.917	400	0.133	0.105
germany	randomwalknowcast	wis	DE-BY	all	27.5	0.0218	0.0895	0.477	0.885	400	0.118	0.108
germany	randomwalknowcast	wis	DE-HB	all	4.21	0.0493	-0.0495	0.562	0.825	400	0.663	0.479
germany	randomwalknowcast	wis	DE-HE	all	10.8	0.0231	-0.386	0.468	0.892	400	0.149	0.124
germany	randomwalknowcast	wis	DE-HH	all	7.82	0.0453	-0.114	0.527	0.885	400	0.236	0.173
germany	randomwalknowcast	wis	DE-MV	all	5.83	0.0273	-0.0867	0.552	0.938	400	0.188	0.155
germany	randomwalknowcast	wis	DE-NI	all	6.99	0.0202	-0.113	0.595	0.895	400	0.151	0.128

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	randomwalknowcast	wis	DE-NW	all	33.8	0.0244	-0.275	0.438	0.958	400	0.138	0.112
germany	randomwalknowcast	wis	DE-RP	all	11.5	0.0363	-0.0634	0.51	0.84	400	0.187	0.152
germany	randomwalknowcast	wis	DE-SH	all	4.41	0.025	-0.402	0.583	0.86	400	0.165	0.153
germany	randomwalknowcast	wis	DE-SL	all	1.48	0.0272	-0.0555	0.63	0.915	400	0.261	0.26
germany	randomwalknowcast	wis	DE-SN	all	44.3	0.0729	-0.0766	0.525	0.943	400	0.321	0.262
germany	randomwalknowcast	wis	DE-ST	all	15.7	0.0463	-0.268	0.515	0.85	400	0.253	0.214
germany	randomwalknowcast	wis	DE-TH	all	9.3	0.0223	0.0341	0.505	0.932	400	0.147	0.147

Ebola hub submissions

Table 7: Ebola hub submissions scores by horizon. Illustrative, dummy data

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	BDBVHosp-baseline	crps	-3	1.06	0.116	-0.0412	0.759	0.981	108	1	1

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	BDBVHorps baseline	crps	-2	1.08	0.113	-0.0332	0.824	0.981	108	1	1
ebola	BDBVHorps baseline	crps	-1	1.48	0.157	-0.0709	0.778	0.981	108	1	1
ebola	BDBVHorps baseline	crps	0	2.78	0.291	-0.0121	0.639	0.954	108	1	1
ebola	BDBVHorps baseline	crps	1	3.64	0.308	0.0156	0.735	0.961	102	1	1
ebola	BDBVHorps baseline	crps	2	5.44	0.44	0.00839	0.583	0.938	96	1	1
ebola	BDBVHorps baseline	crps	3	7.19	0.405	0.0537	0.756	1	90	1	1
ebola	BDBVHorps baseline	crps	4	13.3	0.6	0.0888	0.589	0.911	90	1	1
ebola	fm_chromplete	crps_weekly_com	-3	1.12	0.13	-0.251	0.685	0.889	108	1.05	1.12
ebola	fm_chromplete	crps_weekly_com	-2	1.24	0.142	-0.197	0.63	0.917	108	1.15	1.25
ebola	fm_chromplete	crps_weekly_com	-1	1.95	0.217	-0.148	0.556	0.861	108	1.32	1.38
ebola	fm_chromplete	crps_weekly_com	0	3.14	0.309	-0.0892	0.509	0.861	108	1.13	1.06
ebola	fm_chromplete	crps_weekly_com	1	4.03	0.401	-0.0824	0.539	0.882	102	1.11	1.3
ebola	fm_chromplete	crps_weekly_com	2	4.96	0.44	-0.0562	0.562	0.906	96	0.911	0.999
ebola	fm_chromplete	crps_weekly_com	3	9.15	0.599	0.0456	0.433	0.878	90	1.27	1.48
ebola	fm_chromplete	crps_weekly_com	4	22.8	0.644	0.0175	0.556	0.9	90	1.71	1.07
ebola	fm_chrom	crps_weekly_ragg	-3	1.07	0.123	-0.2	0.62	0.833	108	1.01	1.06
ebola	fm_chrom	crps_weekly_ragg	-2	1.23	0.134	-0.00838	0.593	0.87	108	1.14	1.18
ebola	fm_chrom	crps_weekly_ragg	-1	1.53	0.149	-0.0729	0.602	0.898	108	1.04	0.946
ebola	fm_chrom	crps_weekly_ragg	0	2.58	0.245	0.00569	0.537	0.815	108	0.925	0.842
ebola	fm_chrom	crps_weekly_ragg	1	3.76	0.369	-0.0302	0.441	0.804	102	1.03	1.2
ebola	fm_chrom	crps_weekly_ragg	2	6.08	0.443	-0.0204	0.479	0.76	96	1.28	1.01

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	fm_chronos_weekly_ragged	crps	-3	5.99	0.421	0.0312	0.556	0.844	90	0.833	1.04
ebola	fm_chronos_weekly_ragged	crps	-2	7.24	0.545	0.0256	0.411	0.8	90	0.543	0.908
ebola	fm_chronos_weekly_ragged_ft	crps	-3	0.978	0.105	0.0956	0.667	0.944	108	0.902	0.908
ebola	fm_chronos_weekly_ragged_ft	crps	-2	1.12	0.108	0.237	0.667	0.944	108	1.04	0.954
ebola	fm_chronos_weekly_ragged_ft	crps	-1	1.46	0.141	0.137	0.602	0.972	108	0.991	0.895
ebola	fm_chronos_weekly_ragged_ft	crps	0	2.32	0.192	0.131	0.676	0.944	108	0.832	0.659
ebola	fm_chronos_weekly_ragged_ft	crps	1	3.31	0.267	0.0158	0.618	0.922	102	0.91	0.867
ebola	fm_chronos_weekly_ragged_ft	crps	2	4.03	0.32	0.117	0.583	0.927	96	0.741	0.726
ebola	fm_chronos_weekly_ragged_ft	crps	3	7.11	0.38	0.122	0.644	0.889	90	0.988	0.938
ebola	fm_chronos_weekly_ragged_ft	crps	4	6.6	0.455	0.0624	0.578	0.889	90	0.494	0.758
ebola	hub_ensemble	crps	-3	1.07	0.11	0.202	0.63	0.917	108	1.01	0.947
ebola	hub_ensemble	crps	-2	1.33	0.122	0.171	0.602	0.907	108	1.23	1.08
ebola	hub_ensemble	crps	-1	1.51	0.141	0.114	0.639	0.926	108	1.02	0.893
ebola	hub_ensemble	crps	0	2.43	0.21	0.179	0.63	0.907	108	0.873	0.722
ebola	hub_ensemble	crps	1	3.49	0.262	0.115	0.559	0.892	102	0.961	0.85
ebola	hub_ensemble	crps	2	4.56	0.346	0.0401	0.531	0.854	96	0.839	0.786
ebola	hub_ensemble	crps	3	7.57	0.427	0.203	0.556	0.856	90	1.05	1.06
ebola	hub_ensemble	crps	4	5.39	0.363	0.0838	0.633	0.967	90	0.404	0.605
ebola	hub_teamcrps_a	crps	-3	1.94	0.168	0.605	0.38	0.741	108	1.83	1.45
ebola	hub_teamcrps_a	crps	-2	2.48	0.193	0.605	0.398	0.685	108	2.3	1.7
ebola	hub_teamcrps_a	crps	-1	2.39	0.204	0.449	0.491	0.778	108	1.62	1.3
ebola	hub_teamcrps_a	crps	0	3.89	0.284	0.354	0.444	0.824	108	1.4	0.974
ebola	hub_teamcrps_a	crps	1	5.9	0.388	0.358	0.441	0.775	102	1.62	1.26
ebola	hub_teamcrps_a	crps	2	5.43	0.39	0.194	0.51	0.812	96	0.998	0.887

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	hub_teamcmpsa		3	9.39	0.491	0.272	0.367	0.811	90	1.31	1.21
ebola	hub_teamcmpsa		4	12.8	0.572	0.123	0.422	0.8	90	0.957	0.953
ebola	hub_teamcmpsb		-3	0.915	0.113	-0.18	0.75	0.981	108	0.861	0.981
ebola	hub_teamcmpsb		-2	1.26	0.128	-0.00593	0.722	0.981	108	1.17	1.13
ebola	hub_teamcmpsb		-1	1.42	0.155	0.034	0.741	0.981	108	0.963	0.988
ebola	hub_teamcmpsb		0	2.48	0.257	0.00208	0.639	0.944	108	0.89	0.882
ebola	hub_teamcmpsb		1	3.27	0.309	0.0478	0.667	0.971	102	0.901	1
ebola	hub_teamcmpsb		2	6.09	0.444	0.109	0.521	0.938	96	1.12	1.01
ebola	hub_teamcmpsb		3	6.39	0.44	0.0899	0.656	0.989	90	0.889	1.09
ebola	hub_teamcmpsb		4	7.87	0.597	-0.0296	0.567	0.922	90	0.59	0.995
ebola	random_walk	crps	-3	1.21	0.133	-0.141	0.75	0.935	108	1.14	1.15
ebola	random_walk	crps	-2	1.19	0.134	-0.179	0.722	0.991	108	1.1	1.18
ebola	random_walk	crps	-1	1.79	0.185	-0.169	0.657	0.972	108	1.21	1.17
ebola	random_walk	crps	0	2.6	0.251	-0.0209	0.657	0.991	108	0.932	0.861
ebola	random_walk	crps	1	4.24	0.35	-0.0376	0.598	0.971	102	1.17	1.14
ebola	random_walk	crps	2	4.73	0.397	-0.0362	0.698	0.948	96	0.87	0.902
ebola	random_walk	crps	3	6.83	0.543	-0.106	0.622	0.956	90	0.95	1.34
ebola	random_walk	crps	4	11.7	0.564	0.0386	0.667	0.956	90	0.878	0.939
ebola	random_walk_nowcast	crps	-3	1.08	0.12	0.0284	0.704	0.972	108	1.02	1.04
ebola	random_walk_nowcast	crps	-2	1.05	0.116	0.0351	0.815	1	108	0.978	1.03
ebola	random_walk_nowcast	crps	-1	1.52	0.159	0.0807	0.722	0.981	108	1.03	1.01

study	model	metric	horizon	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	randomwalknowcast	crps	0	2.2	0.202	0.083	0.778	0.972	108	0.789	0.692
ebola	randomwalknowcast	crps	1	3.41	0.304	0.131	0.706	0.98	102	0.937	0.986
ebola	randomwalknowcast	crps	2	4.61	0.345	0.0479	0.719	0.979	96	0.849	0.785
ebola	randomwalknowcast	crps	3	5.35	0.398	0.0416	0.7	0.989	90	0.743	0.982
ebola	randomwalknowcast	crps	4	6.85	0.423	0.0758	0.744	0.989	90	0.513	0.705

Table 8: Ebola hub submissions scores by location and stratum. Illustrative, dummy data

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	BDBVHubbaseline	crps	loc01	all	2.58	0.228	0.0282	0.733	0.978	135	1	1
ebola	BDBVHubbaseline	crps	loc02	all	9.34	0.312	0.0321	0.681	0.956	135	1	1
ebola	BDBVHubbaseline	crps	loc03	all	2.69	0.266	-0.055	0.696	0.97	135	1	1
ebola	BDBVHubbaseline	crps	loc04	all	3.77	0.309	-0.0163	0.733	0.963	135	1	1
ebola	BDBVHubbaseline	crps	loc05	all	2.44	0.322	0.014	0.719	0.956	135	1	1
ebola	BDBVHubbaseline	crps	loc06	all	4.6	0.321	-0.016	0.704	0.963	135	1	1

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	fm_chronos_weekly_complete	rps	loc01	all	3.4	0.319	-0.077	0.578	0.919	135	1.32	1.4
ebola	fm_chronos_weekly_complete	rps	loc02	all	5.91	0.277	-0.107	0.541	0.874	135	0.632	0.889
ebola	fm_chronos_weekly_complete	rps	loc03	all	14.7	0.475	-0.0693	0.496	0.837	135	5.47	1.78
ebola	fm_chronos_weekly_complete	rps	loc04	all	4.69	0.382	-0.0852	0.593	0.904	135	1.24	1.24
ebola	fm_chronos_weekly_complete	rps	loc05	all	1.92	0.31	-0.096	0.63	0.941	135	0.787	0.964
ebola	fm_chronos_weekly_complete	rps	loc06	all	3.17	0.319	-0.174	0.533	0.844	135	0.689	0.993
ebola	fm_chronos_weekly_ragged	rps	loc01	all	3.75	0.309	0.104	0.481	0.837	135	1.45	1.36
ebola	fm_chronos_weekly_ragged	rps	loc02	all	7.03	0.249	-0.0271	0.526	0.822	135	0.752	0.798
ebola	fm_chronos_weekly_ragged	rps	loc03	all	2.92	0.295	-0.0681	0.593	0.83	135	1.09	1.11
ebola	fm_chronos_weekly_ragged	rps	loc04	all	2.78	0.267	-0.0933	0.585	0.859	135	0.738	0.864
ebola	fm_chronos_weekly_ragged	rps	loc05	all	1.89	0.318	-0.0384	0.526	0.807	135	0.776	0.988
ebola	fm_chronos_weekly_ragged	rps	loc06	all	3.38	0.32	-0.0974	0.489	0.822	135	0.734	0.998
ebola	fm_chronos_weekly_ragged_ft	rps	loc01	all	2.82	0.245	0.166	0.57	0.911	135	1.09	1.07
ebola	fm_chronos_weekly_ragged_ft	rps	loc02	all	7.62	0.276	0.0902	0.548	0.896	135	0.816	0.886
ebola	fm_chronos_weekly_ragged_ft	rps	loc03	all	2.03	0.223	0.113	0.637	0.978	135	0.754	0.839
ebola	fm_chronos_weekly_ragged_ft	rps	loc04	all	2.4	0.213	0.0893	0.644	0.956	135	0.637	0.688
ebola	fm_chronos_weekly_ragged_ft	rps	loc05	all	1.26	0.218	0.136	0.733	0.941	135	0.517	0.678
ebola	fm_chronos_weekly_ragged_ft	rps	loc06	all	3.06	0.247	0.104	0.652	0.904	135	0.665	0.77
ebola	hub_ensemble	rps	loc01	all	2.63	0.225	0.109	0.622	0.896	135	1.02	0.99
ebola	hub_ensemble	rps	loc02	all	6.79	0.249	0.197	0.533	0.859	135	0.726	0.798
ebola	hub_ensemble	rps	loc03	all	2.63	0.25	0.131	0.637	0.919	135	0.977	0.94

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	hub_ensemble	crps	loc04	all	2.71	0.237	0.112	0.578	0.933	135	0.717	0.766
ebola	hub_ensemble	crps	loc05	all	1.68	0.247	0.16	0.689	0.948	135	0.69	0.769
ebola	hub_ensemble	crps	loc06	all	3.16	0.228	0.131	0.533	0.867	135	0.687	0.711
ebola	hub_teams_a	crps	loc01	all	3.56	0.315	0.408	0.385	0.726	135	1.38	1.38
ebola	hub_teams_a	crps	loc02	all	10.5	0.335	0.36	0.4	0.778	135	1.12	1.07
ebola	hub_teams_a	crps	loc03	all	3.89	0.314	0.361	0.385	0.793	135	1.45	1.18
ebola	hub_teams_a	crps	loc04	all	5.26	0.334	0.379	0.519	0.778	135	1.39	1.08
ebola	hub_teams_a	crps	loc05	all	3.95	0.368	0.357	0.467	0.815	135	1.62	1.14
ebola	hub_teams_a	crps	loc06	all	4.53	0.294	0.417	0.437	0.77	135	0.984	0.915
ebola	hub_teams_b	crps	loc01	all	3	0.289	0.0593	0.615	0.933	135	1.16	1.27
ebola	hub_teams_b	crps	loc02	all	5.84	0.248	0.0129	0.652	0.963	135	0.625	0.795
ebola	hub_teams_b	crps	loc03	all	3.35	0.324	-0.023	0.593	0.97	135	1.25	1.22
ebola	hub_teams_b	crps	loc04	all	3.38	0.312	-0.0105	0.659	0.978	135	0.895	1.01
ebola	hub_teams_b	crps	loc05	all	2.67	0.344	-0.0201	0.763	0.978	135	1.1	1.07
ebola	hub_teams_b	crps	loc06	all	2.93	0.245	0.0156	0.689	0.963	135	0.637	0.764
ebola	random_walk	crps	loc01	all	3.16	0.261	-0.0745	0.711	0.97	135	1.23	1.15
ebola	random_walk	crps	loc02	all	8.64	0.293	-0.0773	0.622	0.948	135	0.925	0.939
ebola	random_walk	crps	loc03	all	2.57	0.314	-0.131	0.689	0.985	135	0.956	1.18
ebola	random_walk	crps	loc04	all	4.46	0.34	-0.0299	0.659	0.963	135	1.18	1.1
ebola	random_walk	crps	loc05	all	2.1	0.291	-0.0214	0.704	0.963	135	0.862	0.903
ebola	random_walk	crps	loc06	all	3.42	0.349	-0.172	0.652	0.963	135	0.743	1.09

study	model	metric	location	stratum	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	random-walk-nowcast	crps	loc01	all	3.37	0.267	0.114	0.689	0.941	135	1.31	1.17
ebola	random-walk-nowcast	crps	loc02	all	3.65	0.166	0.0877	0.719	0.993	135	0.39	0.532
ebola	random-walk-nowcast	crps	loc03	all	2.43	0.267	0.0206	0.748	1	135	0.905	1
ebola	random-walk-nowcast	crps	loc04	all	4.77	0.326	0.0246	0.711	0.985	135	1.26	1.05
ebola	random-walk-nowcast	crps	loc05	all	1.69	0.245	0.0875	0.785	0.993	135	0.692	0.762
ebola	random-walk-nowcast	crps	loc06	all	2.76	0.228	0.0581	0.77	0.985	135	0.601	0.712

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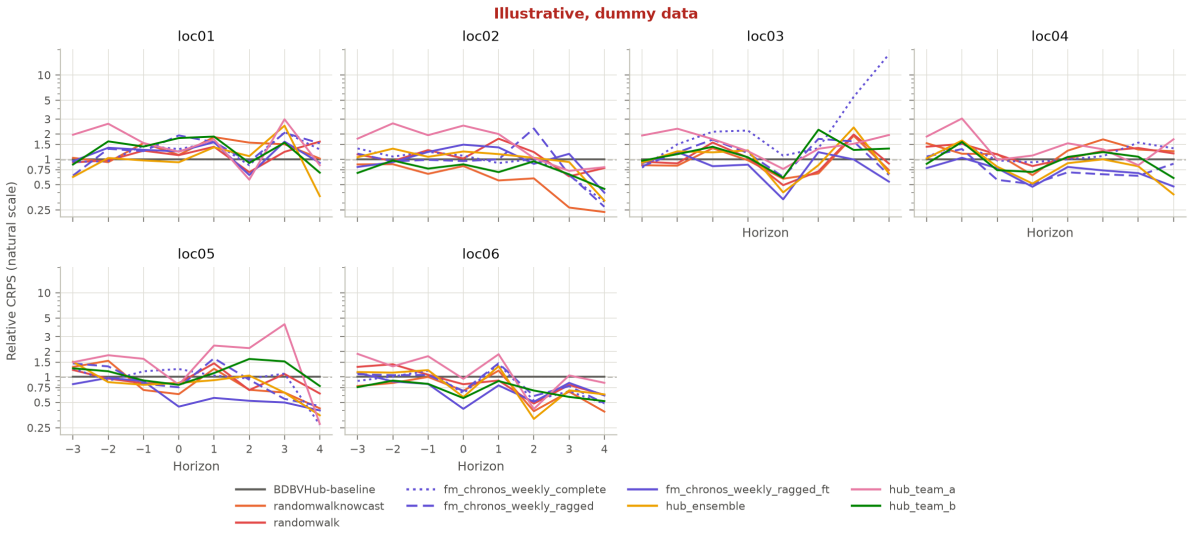


Figure 9: Ebola hub submissions: relative CRPS by location and horizon. Illustrative, dummy data

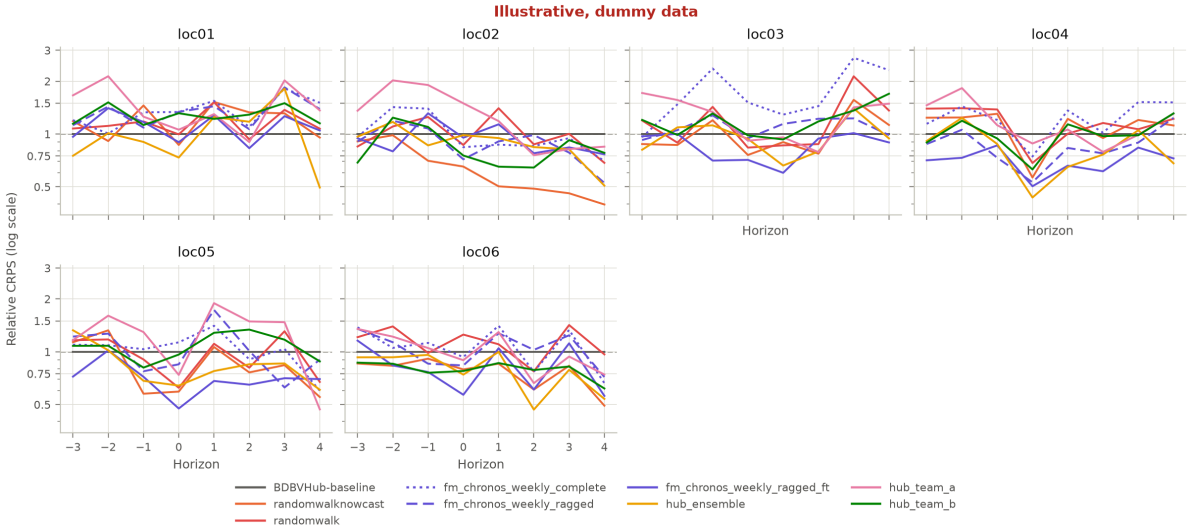


Figure 10: Ebola hub submissions: relative CRPS on the log scale by location and horizon. Illustrative, dummy data

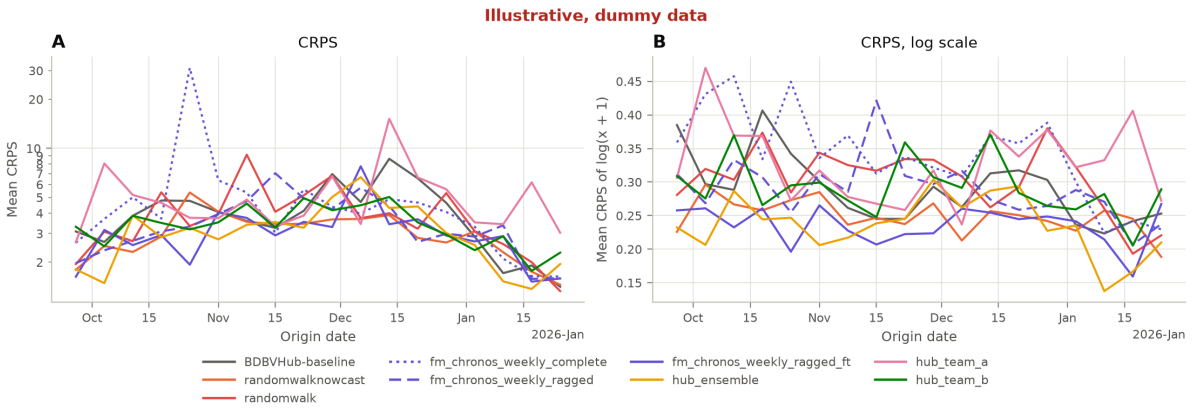


Figure 11: Ebola hub submissions: CRPS over time. Illustrative, dummy data

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